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A stepwise model of Reaction-Diffusion and Positional-Information governs self-organized human peri-gastrulation-like patterning

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Author Contributions

M.T. and P.W.Z conceived and designed all experiments and wrote the manuscript. M.T developed the micro-patterning technology, performed all cell culture, experiments, imaging, and mathematical modeling. M.T. and L.P. performed RNA silencing and in-situ hybridization. M.T., J.O, and T.Z.C performed image analysis. M.T. and N.S. performed gene expression analysis. R.F.G provided critical experimental advice, and oversaw image analysis methodology.

Abstract

How position-dependent cell fate acquisition occurs during embryogenesis is a central question in developmental biology. To study this process, we developed a defined, high-throughput assay to induce peri-gastrulation-associated patterning in geometrically-confined human pluripotent stem cell (hPSC) colonies. We observed that, upon BMP4 treatment, phosphorylated SMAD1 (pSMAD1) activity in the colonies organized into a radial gradient. We developed a Reaction-Diffusion (RD) based computational model and observed that the self-organization of pSMAD1 signaling was consistent with the RD principle. Consequent fate acquisition occurred as a function of both pSMAD1 signaling strength and duration of induction – consistent with the Positional-Information (PI) paradigm. We propose that the self-organized peri-gastrulation-like fate patterning in BMP4 treated geometrically-confined hPSC colonies arises via a stepwise model of RD, and PI. This two-step model predicted experimental responses to perturbations of key parameters such as colony size, and BMP4 dose. Furthermore, it also predicted experimental conditions that resulted in RD-like periodic patterning in large hPSC colonies, and rescued peri-gastrulation-like patterning in colony sizes previously thought to be reticent to this behaviour.

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Introduction

During development, pluripotent stem cells (PSCs) in the epiblast are exposed to signaling gradients that initiate a sequence of fate specifications and cell movements resulting in spatial segregation of the germ layers in a developmentally conserved process called gastrulation (Rossant & Tam 2004; Rossant & Tam 2009; Morris et al. 2012; Tam et al. 2006; Ciruna & Rossant 2001). A number of model organisms have been used to study the molecular mechanisms that underpin the fate patterning that arises during this critical developmental checkpoint (Solnica-Krezel & Sepich 2012; Leptin 2005; Keller 2005; Nakaya & Sheng 2008; Tam & Behringer 1997; Tam & Loebel 2007; Myers et al. 2002). In all model organisms studied, transforming growth factor beta (TGFβ) superfamily members including bone morphogenetic proteins (BMPs) and Nodal, which signal though a family of mediator proteins called SMADs, play key roles in inducing gastrulation-specific pattern formation (Wu & Hill 2009). Although involvement of the TGFβ superfamily members is conserved between species during gastrulation, details of fate patterning in relation to cell movements, tissue structure, specific molecules involved etc., can vary between species (Wu & Hill 2009; Rossant 2015; Narasimha & Leptin 2000; Solnica-Krezel & Sepich 2012; Solnica-Krezel 2005). Therefore, it has been challenging to relate the regulatory mechanisms of pattern formation in previously studied developmental models to human gastrulation.

Since human embryos are not typically available for direct investigation, studying human gastrulation-associated fate patterning requires *in vitro* platforms that allow robust simulation and investigation of the signaling programs that initiate and drive gastrulation-like events. We and others have previously used micro-patterning technologies to control human (h)PSC colony geometry, demonstrating improved cell response consistency than is achieved in conventional adherent cultures (Nazareth et al. 2013; Peerani et al. 2007; Bauwens et al. 2008; Bauwens et al. 2011; Alom Ruiz & Chen 2007; Stevens et al. 2013; Ma et al. 2015). These studies highlight that control of endogenous signaling profiles, cell-cell contact, and mechanical forces are crucial to robustly regulate cell fate and spatial tissue organization. Recently, Warmflash *et al*. (Warmflash et al. 2014) used similar techniques to demonstrate that following BMP4 treatment, geometrically controlled hPSC colonies recapitulate many aspects of the peri-gastrulation-stage epiblast. Specifically, these colonies exhibited spatially patterned regions characteristic of primitive-streak-like, mesoderm-like, endoderm-like, ectoderm-like and trophoblast-like tissues (Warmflash et al. 2014).

Two prominent biochemical models have influenced our understanding of cell fate patterning and morphogenesis that occurs during embryogenesis. The first is Reaction-Diffusion (RD), which describes the self-organization of homogenously distributed signaling molecules (morphogens) into complex, asymmetric patterns that provide spatial information to developing tissues (Turing 1952; Gierer & Meinhardt 1972). The second is Positional Information (PI), which describes how the asymmetric morphogen distributions across a developing tissue can be interpreted, and result in cell fate patterning (Wolpert 1981; Wolpert 1969). RD hypothesizes the presence of an interaction network of two molecules, an ‘activator’ which activates the expression of both molecules, and an ‘inhibitor’ which inhibits their expression. This interaction network, in conjunction with dissimilar activator and inhibitor diffusivities, is theoretically sufficient to self-organize asymmetrical morphogen distributions (Turing 1952; Gierer & Meinhardt 1972). The initial version of PI proposed a mechanism by which this asymmetric morphogen distribution could be translated into patterned developmental fates through a signaling threshold-based mechanism. Subsequent studies using the developing neural tube have demonstrated that fate patterning by PI is not just mediated by morphogen threshold levels, but is a function of both the morphogen concentration and exposure duration (Dessaud et al. 2007; Dessaud et al. 2008; Briscoe & Small 2015).

Here we demonstrate, using fully defined and scalable conditions, that geometrically-confined hPSC colonies organize into radially segregated regions that express markers characteristic of ectoderm-like, primitive streak-like, and trophoblast-like tissues. We show that upon BMP4 induction, an RD network regulated by BMP4 and NOGGIN (a cardinal BMP inhibitor) rapidly organizes nuclear localized phosphorylated (p)SMAD1 (effector of BMP signaling) into a gradient within the geometrically-confined

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colonies. The established gradient then patterns hPSC differentiation in a manner consistent with PI. We developed a computational model of a BMP4-NOGGIN RD system and demonstrate that, across a range of colony sizes and BMP4 doses, RD consistently predicts the formation of pSMAD1 signaling gradient and PI accurately predicts the patterned fate acquisition. A stepwise model of RD mediated self-organization of pSMAD1 signaling gradient followed by a PI mediated patterning of cell fates can predict the outcome of previously unexplored experimental conditions. Specifically, we both predict and observe periodic fate patterning consistent with the cardinal RD paradigm. Furthermore, our model can identify conditions that rescue fate patterning in colonies previously deemed incapable of facilitating pattern formation (Warmflash et al. 2014). Taken together, our data support the concept that a stepwise process of RD and PI controls the peri-gastrulation-like patterns that develop in differentiating hPSC colonies.

Results

A defined high throughput assay for induction of peri-gastrulation-like patterning in human pluripotent stem cell colonies

Consistent with recent reports (Warmflash et al. 2014; Etoc et al. 2016), we observed radially segregated expression of gastrulation-associated markers CDX2, Brachyury (BRA), and SOX2 – representative of trophoblast-like, primitive-streak-like, and ectoderm-like tissues respectively – develop in geometrically-confined circular hPSC colonies differentiated in mouse embryonic fibroblast conditioned medium (CM) supplemented with BMP4 (**Fig. 1A**). Given the challenge of identifying key molecules regulating pattern formation in undefined CM, our first aim was to identify defined basal conditions that induce these patterns in hPSC colonies. Further, we optimized a previously described protocol that uses Deep Ultraviolet light (< 200 nm) mediated photo-oxidation of polyethylene glycol (PEG) coated slides (Azioune et al. 2009) to allow high-fidelity patterning of hPSC colonies and adapted this technique to produce 96-well microtiter plates (Materials and Methods).

Using our PEG plates, we performed a medium screen to identify defined conditions to induce patterning of gastrulation-associated fates in geometrically-confined hPSC colonies. The screen consisted of Nutristem (NS), mTeSR (MT), Essential-8 (E8), a Knockout Serum Replacement based medium (SR), and an N2B27 based medium – all supplemented with BMP4. Preliminary testing with N2B27 medium revealed that NODAL supplementation elicited a positive response in the induction of the BRA expressing region (data not shown), consistent with the importance of Nodal signaling in the specification of the primitive streak fate (Funa et al. 2015; Schier 2003; Brennan et al. 2001; Shen 2007). Therefore, we performed all N2B27 based experiments with supplemented NODAL (100ng/ml). We found that, although segregated expression of CDX2 and SOX2 developed in all defined test media, BRA expression was not consistently observed in E8, NS, and SR media (**Fig. S1, S2**). Further, while spatial segregation of CDX2, BRA, and SOX2 occurred reproducibly in MT medium, colonies patterned in MT tended to lift off the plate by 48h. N2B27 medium robustly gave rise to differentiating hPSC colonies with regions expressing trophoblast-associated (CDX2) and primitive-streak-associated (BRA) markers, with concurrent spatial patterning of expression markers indicative of endoderm-like (SOX17) and mesoderm-like (EOMES) fates, in a manner indistinguishable from hPSC colonies differentiated in BMP4-supplemented CM (**Fig. 1A-C, Fig. S1, S2**). Further, the regions at the center of the differentiating colonies expressed SOX2 (**Fig. 1B**) but not NANOG (**Fig. S3**), indicating a prospective ectoderm-like fate. Notably, the SNAIL-expressing mesenchymal cells appeared underneath the EPCAM-expressing epithelial layer in the regions expressing markers of the primitive streak (**Fig. 1D**), which parallels the organization of the mesenchymal and epithelial cells at the onset of gastrulation. Also, this observation is consistent with what has previously been reported in BMP4-supplemented CM (Warmflash et al. 2014). Given these results, we define the fate patterning observed in BMP4 supplemented N2B27 medium as 'peri-gastrulation-like', and unless otherwise stated, this induction medium was used for all studies detailed below.

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Nodal signaling is necessary for BRA expression but does not induce peri-gastrulation-like patterning

Nodal signaling is important in establishing the primitive streak fate (Hong et al. 2011; Funa et al. 2015). Given that our N2B27 medium contained NODAL, we first examined whether the spatially organized differentiation observed within the geometrically confined hPSC colonies required Nodal signaling. We found that selectively inhibiting Nodal signaling with SB-431542 (SB), an inhibitor of Alk4/5/7 receptors (Inman et al. 2002), in N2B27 medium supplemented with BMP4, abrogated the expression of BRA (**Fig 2A-C**) and significantly reduced the expression of CDX2 (**Fig 2A**). However, patterning of regions expressing SOX2 and residual CDX2 was still maintained (**Fig 2C-E**). These data indicate that Nodal signaling was necessary for the expression of BRA, and at least a subset of the CDX2, but did not induce organized fate patterning in the geometrically confined hPSC colonies. We then tested the necessity of BMP signaling to induce the peri-gastrulation-like fates within the geometrically confined hPSC colonies. To test this, we queried if the patterned fates within the hPSC colonies would arise in either N2B27 medium without supplemented BMP4, or in N2B27+BMP4 with supplementation of a small molecule BMP inhibitor LDN-193189 (LDN) (Cuny et al. 2008). We found that in both cases there was no observed peri-gastrulation-like patterning (**Fig 2A-B**). Furthermore, this patterning deficiency in N2B27 without supplemented BMP4 could not be rescued by further addition of either 100ng/ml or 200ng/ml of NODAL (**Fig 2A-B**). These data indicated that activation of BMP signaling was necessary to induce peri-gastrulation-like differentiation within the geometrically confined hPSC colonies. Taken together these data indicate that induction of peri-gastrulation-like fate patterning requires BMP signaling and the formation of BRA-expression region further requires Nodal activity.

BMP4-NOGGIN interaction network regulates pSMAD1 gradient self-organization

Since we observed that BMP signaling was necessary for organized fate patterning within the geometrically confined hPSC colonies, we next examined how downstream effectors of BMP signaling were organized within the colonies. BMP ligands activate BMP receptors (BMPR1A, BMPR1B, BMPR2) leading to phosphorylation and nuclear translocation of SMAD1 followed by the transcription of context-specific target genes (Zhang & Li 2005). We used an antibody specific to pSMAD1 to measure the distribution of nuclear localized SMAD1 activity as a function of radial distance from colony centers at 1, 6, 18, and 24h following BMP4 induction. Robust analysis was enabled by overlaying pSMAD1 expression profiles of at least 100 colonies at each time point, yielding the average intensity of pSMAD1 as a function of colony radius. We noted that 1h after BMP4 induction, although the pSMAD1 activity appeared at all colony radii, the expression on average was slightly lower at the colony center then at the periphery (**Fig. 3A**), suggesting that centers of hPSC colonies might contain BMP inhibitors. This is consistent with previous reports that in the pluripotent state, hPSCs express secreted factors such as FOLLISTATIN, CERL, GDF3 etc. (Vallier et al. 2004; Yang et al. 2015; Besser 2004). Although these factors belong to the Activin/Nodal family, they are competent inhibitors of BMP signaling (Wu & Hill 2009). We observed relatively higher expression levels of these Activin/Nodal family member inhibitors of BMP signaling than the canonical BMP inhibitors like NOGGIN, and CHORDIN in hPSCs during regular culture conditions (**Fig. S4**). The secretion of these BMP inhibitors in the geometrically-confined hPSC colony would produce a spatial distribution with the center of the colonies having relatively higher concentrations of BMP inhibitors (**Supplement – Model description**, **Fig. S14**). In a colony with this expression profile of non-canonical BMP inhibitors, BMP4 treatment would result in a small reduction of BMP signaling activity at the center of the colony, as we observed (**Fig. 3A**). Subsequently, over the first 24h of BMP4 treatment, we observed a rapid downregulation of pSMAD1 activity at the colony centers, and spontaneous organization into a radial gradient with the cells in the periphery exhibiting higher levels of pSMAD1 activity (**Fig. 3A-C**). To test whether the self-organization of pSMAD1 activity could be attributed to regulatory feedback of the BMP pathway, we measured the

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expression of both positive and negative feedback mediators of BMP signaling over the first 26h after BMP4 treatment. We found significant upregulation in the expression of both NOGGIN and BMP4 (**Fig. 3D**) in a dose dependent manner (**Fig. S5**). Notably, BMP4 and NOGGIN upregulation was seen in all basal medium conditions tested (**Fig. S6**), indicating that the upregulation was a response to the activation of BMP signaling and unrelated to the N2B27 medium. Furthermore, NOGGIN knockdown, using siRNA, significantly increased pSMAD1 activity in the central regions of the geometrically-confined colonies demonstrating a role for NOGGIN in the self-organization of the pSMAD1 gradient (**Fig. 3E-G, Fig. S7**), as has recently been proposed by others (Etoc et al. 2016). Interestingly, along with the NOGGIN mediated negative feedback, we noted the presence of a positive feedback loop in BMP signaling wherein BMP4 supplementation resulted in BMP4 expression (**Fig. 3D, Fig. S6**). Together, the presence of both a positive and negative feedback response mediated by BMP4 suggested the presence of a BMP4-NOGGIN RD network underlying the pSMAD1 radial self-organization in differentiating hPSC colonies (**Fig. 3H**). We next set out to investigate properties of the RD framework as they relate to the peri-gastrulation-like patterning.

pSMAD1 gradient formation is colony size and BMP4 concentration dependent

To simulate RD mediated self-organization of pSMAD1 activity, we developed a finite element model that predicts the spatio-temporal distribution of signaling-competent, free BMP4 ligands within the differentiating geometrically confined colonies using the RD-specific two-component, coupled, partial differentiation equation set (Turing 1952; Gierer & Meinhardt 1972) (**Supplement – Model Description, Movie 1**). To determine if this model could accurately predict experimental data, we performed sweeps on two key model parameters: the initial concentration of BMP4 in the induction medium (BMPi), and the colony size. Our model predicted that reducing BMPi while maintaining the colony diameter at 1000μm would still lead to the formation of distribution gradients of free BMP4 within the colonies, but with lower ligand levels at the colony periphery (**Fig. 4A-B**). Consistent with model predictions, the spatial distribution of the pSMAD1 concentration in 1000μm diameter colonies 24h after induction with varying BMPi (6.25ng/ml, 12.5ng/ml, 25ng/ml, and 50ng/ml) formed gradients, with lower BMPi conditions producing lower levels of peak pSMAD1 activity at colony peripheries (**Fig. 4C-E, Fig. S8**). We next queried the model to predict how reducing colony size, at a fixed BMPi dose, would affect the spatial distribution of free BMP4 ligands in the differentiating colonies. The model predicted that reducing the colony size would progressively increase the presence of free BMP4 ligands at the colony centers (**Fig. 4F-G**). To test the model predictions experimentally, we differentiated colonies of varying sizes with constant BMPi dose (50ng/ml), and assessed pSMAD1 activity 24h post-induction. Consistent with the model predictions, we found that small colonies were unable to form regions with low pSMAD1 activity (**Fig. 4H-I**). These findings are consistent with our hypothesis that the self-organization of pSMAD1 activity observed in differentiating hPSC colonies is governed by a BMP4-NOGGIN RD network.

Peri-gastrulation-like fates arise in a manner consistent with the PI paradigm

Notably, the perturbations we tested above resulted in a change in pSMAD1 activity both at the periphery (**Fig. 4D**) and at the center (**Fig. 4I**) of the colonies. We reasoned that if fate acquisition in these colonies were a function of pSMAD1 activity thresholds, we would observe fate switches at different conditions. Accordingly, we tested the same conditions above – this time after 48h – and stained for the fate-associated markers SOX2, BRA, and CDX2. First, to perturb the pSMAD1 activity at the colony periphery we varied BMPi doses while keeping the colony size constant (**Fig. 5A**). Consistent with a threshold-dependent fate acquisition model, we found a significant reduction of CDX2 expression in colonies induced at low BMPi conditions (**Fig. 5B-C**). Furthermore, when we perturbed the pSMAD1 activity at the colony center by varying colony size at a constant BMPi dose (**Fig. 5D**), we found that SOX2 expression disappeared in smaller colonies ($\leq$ 300μm diameter) (**Fig. 5E-F**). These data indicate that BMP signaling thresholds dictate fate acquisition in the differentiating geometrically-confined hPSC colonies.

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The current understanding of PI, however, suggests that transcription factors associated with patterned fates arise not just as a function of the morphogen concentration, but also as a function of induction time; i.e. the fates associated with higher levels of morphogen concentration are induced after longer induction times (Briscoe & Small 2015; Green & Sharpe 2015; Dessaud et al. 2008; Dessaud et al. 2007). Therefore, we set out to investigate if fates in the differentiating hPSC colonies arose as a function of both the level of pSMAD1 activity and the time of induction. Accordingly, we tested four different BMPi doses (50ng/mL, 25ng/mL, 12.5ng/mL, and 6.25ng/mL), and analyzed the patterned fates that emerged at four different induction times (12h, 24h, 36h, and 48h). We found that the BRA and CDX2 fates did not arise consistently at either lower BMPi doses or at shorter durations of higher BMPi doses (**Fig. 5G-H, and Fig. S9**). This analysis suggests that the cell fate patterning in the hPSC colonies mediated by the pSMAD1 gradient follows PI, as reflected by the characteristic PI-like profile (**Fig. 5I**). In summary, our findings demonstrate that the fate patterning within the differentiating hPSC colonies occurs via the cardinal PI model (**Fig. 5J**).

A two-step process of RD and PI governs peri-gastrulation-like fate patterning

Our data indicate that (1) the formation of the pSMAD1 gradient follows an RD mechanism, and (2) the subsequent fate acquisition follows a PI mechanism, suggesting a two-step process of biological fate patterning in geometrically-confined hPSC colonies. Recognizing that the classic RD models have periodic peaks of signaling activity, and patterned fates (Green & Sharpe 2015; Turing 1952; Kondo & Miura 2010), we queried our mathematical model to identify conditions that would result in a periodic distribution of free BMP ligands. Our model predicted that an increase in the colony size alone would be insufficient to induce this periodic distribution (**Fig. 6A, Movie 2**). Consistent with this prediction, we did not detect a noticeable periodic response in either pSMAD1 levels at 24h post differentiation with BMP4 (**Fig 6B, Fig. S10A**), or BRA expression after 48h of BMP4 treatment with 50ng/ml (**Fig 6C, Fig. S11A**) when we increased colony diameter from 1000μm to 3mm. However, our model predicted that a concomitant increase in the BMP4 dose along with an increase in the colony size would result in the induction of a periodic response of BMP activity (**Fig. 6D, Movie 3**). Consistent with this prediction, we found that differentiating hPSC colonies of 3mm diameter with a BMP4 dose of 200ng/ml resulted in periodicity in both pSMAD1 at 24h (**Fig. 6E, Fig. S10B**), and BRA expression at 48h post induction (**Fig. 6F, Fig. S11B**). We quantitatively compared the theoretical RD-like periodicity with the experimentally observed behavior to assess the predictive ability of the model. We measured the profile of the predicted BMP activity radially from the center of the colony every 30 degrees, for the condition where a 3mm colony was induced to differentiate in presence of 200ng/ml of BMP4, and quantified the dominant periods using a 1D Fourier transform. The same analysis was performed for pSMAD1 and BRA expression patterns (**Fig. S12**) (Materials and Methods). The means and distributions of the theoretically predicted, and experimentally observed periods were not significantly different from each other (Mann-Whitney U: $p>0.5$ and >0.2, and Kolmogorov-Smirnov: $p=0.3781$ and 0.1452 for pSMAD1 and BRA respectively) (**Fig. 6G-I**), validating the ability of the RD model to predict the experimental RD-like periodic response.

As a final validation, we considered the recently reported claims that 'edge-sensing' underlies the observed pattern formation (Etoc et al. 2016). Edge-sensing is a mechanism in which fates are patterned sequentially with the trophectoderm-like fate at the colony edge, followed by the primitive-streak-like region, and the ectoderm-like region positioned at the colony center. Consequently, small colonies of 250μm diameter, that don't have sufficient space from the colony periphery to pattern the primitive-streak-like, and ectoderm-like regions, would be incapable of fate patterning – only allowing the induction of the trophectoderm-like region (Warmflash et al. 2014). Since our interpretation of the data mechanistically implicated a stepwise coordination of RD and PI, our interpretation of the inability of 250μm diameter colonies to induce all three fates differs from the edge-sensing explanation. Specifically, we argue that at a BMPi dose of 50ng/ml (the dose tested in the previous reports), RD mediated organization of free BMP4 ligands within colonies of 250μm diameter are sustained at high levels throughout the colony (**Fig. 4F-I**), and past the thresholds that

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would induce the primitive-streak-like and ectoderm-like fates after a 48h induction (**Fig. 5E-F**) as per PI. To demonstrate this claim, we queried our model to test if perturbing BMPi could induce the organization of BMP activity at appropriate levels to rescue the fate patterning of all three lineages. Our model predicted that reducing BMPi would reduce the levels of free BMP4 ligands throughout the colony whereby the BRA, and SOX2 expressing regions might be rescued (**Fig. 7A-B**). We tested the expression profiles of pSMAD1 activity at 24h post induction with varying BMPi conditions (50ng/ml, 25ng/ml, 12.5ng/ml, and 6.25ng/ml) of over 500 per dose condition. Consistent with our model predictions, we observed a reduction of pSMAD1 levels overall, but especially in the center of the colonies (**Fig. 7C-E**). These data suggested that lower BMPi conditions might rescue the formation of the BRA, and SOX2 expressing regions in colonies of 250μm diameter in accordance with PI. Indeed, when we tested the fate expression within 250μm diameter colonies after 48h of induction with varying BMPi doses, we observed a significant reduction of CDX2 expression at reduced doses of BMP4, and a concomitant increase of BRA and SOX2 expression within the colonies (**Fig. 7F-G**). Importantly, at a BMPi dose of 12.5ng/ml, patterning of all three fates is rescued in 250μm diameter colonies (**Fig. 7G**).

Taken together, these data are consistent with our hypothesis that the fate acquisition in the geometrically confined hPSC colonies, in response to BMP4 mediated differentiation, arises via a two-step process of RD and PI (**Fig. 8**).

Discussion

Experimental models to study developmental fate patterning

Of the various proposed models of biological pattern formation, RD and PI have emerged as the dominantly accepted mechanisms to describe tissue organization during early development. RD describes a mechanism by which a symmetric morphogen distribution across a developing tissue self-organizes into a signaling gradient, and PI explains how this gradient is interpreted by the tissue resulting in patterned cell fates. Although both RD (Economou et al. 2012; Sick et al. 2006; Raspopovic et al. 2014), and PI (Gregor et al. 2007; Gregor et al. 2008; Houchmandzadeh et al. 2002; Jaeger et al. 2004; Green & Smith 1990; Chen et al. 2012) have been widely studied independently of each other, few developmental model systems allow for studying both aspects of biological pattern formation. Here, we introduced such a model which is a defined *in vitro* platform of developmentally relevant fate patterning using hPSCs to enable studies probing the mechanistic underpinnings behind both RD and PI concurrently.

Relevance to recent studies for peri-gastrulation associated fate patterning

Our study closely follows two recent reports where BMP4 supplemented CM induced similar fate patterning in geometrically-confined hPSC colonies as we observed in our work (Warmflash et al. 2014; Etoc et al. 2016). Although we use defined conditions, our observations were remarkably similar to those reports - underscoring the robustness of their findings. Similar to our claims, Etoc *et al*. (Etoc et al. 2016) also proposed that BMP4 induced fate patterning in hPSC colonies occurs due to a self-organized pSMAD1 signaling gradient, and suggested that this self-organization is regulated by coordination of two different mechanisms. First – a negative feedback control of BMP activity mediated by BMP-signaling induced NOGGIN expression; and second – a density-dependent reduction in sensitivity to BMP4 ligands over the entire colony except for the periphery due to re-localization of BMP receptors from apical to basolateral regions (a mechanism called ‘edge-sensing’), rendering them inaccessible and therefore incapable of inducing the BMP signaling pathway. Our interpretation of both our own data and that reported by Etoc *et al.*, however, differs in a few important aspects outlined below.

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Although their model of dual negative feedback on BMP signaling mediated by NOGGIN and receptor inaccessibility (**Fig. S13A**) predicts the formation of a radial signaling gradient, it does not describe a classical RD system. This is because it lacks positive feedback of the morphogen (Turing 1952; Kondo & Miura 2010; Murray 2008; Gierer & Meinhardt 1972), and would therefore never give rise to periodic RD-like patterns (simplified model shown in **Fig. S13B,C**). In contrast, we demonstrate that the presence of BMP4 in our cultures, regardless of the basal medium used, elicits both a positive, and a negative feedback response mediated by the upregulation of BMP4, and NOGGIN respectively. BMP4 and NOGGIN form a stereotypical activator-inhibitor pair (Gierer & Meinhardt 1972), as BMP4 can induce short range activation of BMP signaling (Zhang & Li 2005), while NOGGIN is a highly-diffusible molecule that can result in long-range inhibition of BMP signaling (Smith & Harland 1992; Inomata et al. 2013). Importantly, we explicitly demonstrate the induction of an RD-like response in larger colonies upon high dose induction of BMP4, both in pSMAD1 activity at 24h after induction and in the fate acquisition as shown by BRA expression 48h after induction (**Fig. 6, S10, S11**).

Etoc *et al*. demonstrate that as cell density increases, the BMP receptors re-localize to basolateral regions of the hPSCs. They also demonstrate that these lateralized receptors are incapable of inducing the BMP signaling pathway in hPSCs when BMP4 ligands are presented on the apical side (Etoc et al. 2016). They claim that this mechanism is one of the drivers of the self-organized formation of the pSMAD1 gradient, given that the cell density of the entire colony increases with time. However, they only investigated this at a 500μm colony diameter. Here, we explored a range of colony sizes, and in 1000μm diameter colonies we generally observed regions of varying cell densities arise 48h after BMP4 induction (**Sup Fig. 1E-F, Fig 2C-D**). This finding is somewhat inconsistent with the mechanism proposed by Etoc *et al*. that the self-organization of a pSMAD1 signaling gradient is induced by density dependent changes that block BMP signaling. An alternate interpretation to the one put forth by Etoc *et al.*, is that the density dependent receptor re-localization is in fact a response to the RD-mediated gradient formation, rather than its cause. This interpretation is directly supported by the appearance of RD-like periodic patterns in both pSMAD1, and BRA expression (**Fig. 6, S10**) in colonies of 3mm diameter where many sporadic spots of higher density relative to the adjacent regions can be observed. In this case, differentiating hPSCs either migrate down a pSMAD1 gradient, or proliferate at different rates depending on the pSMAD1 activity they are exposed to, resulting in regions of varying cell densities. Further work is required to determine which of these possibilities is dominant.

Although our interpretations differ on how 'edge-sensing' mediated receptor availability relates to pSMAD1 gradient formation, the 'edge-sensing' model proposed by Etoc *el al*. (Etoc et al. 2016) nicely parallels our assumed boundary condition for the RD model. For our mathematical RD model, we assumed that the distribution of free BMP4 ligands across the colony evolves over time and space with a fixed concentration of BMP4 at the colony periphery that is equal to the concentration in the bulk medium, a boundary condition that changes depending on the BMP4 concentration of the induction medium. In a recent study, Warmflash, *et al*., claim that the fates are patterned inward from the edge of the geometrically-confined colony (Warmflash et al. 2014). Consequently, they contend that 250μm diameter smaller colonies are incapable of inducing the fate patterning observed in 1000μm diameter colonies, since the smaller colonies do not have enough space to pattern the primitive-streak-like and ectoderm-like regions. However, our interpretation of the 'edge-sensing' model identifies conditions that rescue fate patterning of all three fates in 250μm diameter colonies. Specifically, we demonstrate that in conditions with lower doses of BMP4 in the induction medium, smaller colonies maintain regions that express primitive-streak-like, and ectoderm-like fates after 48h of peri-gastrulation-like induction.

Positional-Information

We specifically tested and demonstrated that, consistent with the PI paradigm, in the differentiating hPSC colonies fate acquisition arises as a function of both pSMAD1 concentration and induction duration. Notably, although the fate acquisition mediated by the pSMAD1 gradient follows the PI paradigm, the primitive-

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streak-like region is not specified in the absence of Nodal signaling. This underscores that patterning of developing tissues depends on coordination of multiple signaling pathways, reminiscent of quintessential models of PI mediated fate patterning such as the gap genes in the *Drosophila* which arise from a collaborative stimulation of multiple signals that include Bicoid, Caudal, Hunchback etc. (Briscoe & Small 2015). Consequently, we propose that although pSMAD1 activity patterns fates according to PI, it does so in conjunction with other transcription factors, for instance SMAD2 which is the effector of Nodal signaling.

Scaling of morphogen gradients during development

Morphogen gradients formed in developing embryos are thought to scale with size. For instance, the scaling of the dorsal-ventral axis in the developing *Xenopus* is robust to dramatic manipulations. In the case where the *Xenopus* embryo undergoes resection of the ventral half, it still develops into a smaller but proportionally patterned larva (Ben-Zvi et al. 2011). This characteristic has been attributed to the function of the Spemann's organizer (Inomata et al. 2013; Ben-Zvi et al. 2011), underscoring that the robustness of the induction and the stabilization of the morphogen gradients that pattern a developing embryo is the result of a concerted effort by multiple tissues. Our data show that fate organization in differentiating micro-patterned hPSC colonies across varying sizes is not robust to a specific BMP4 concentration. Nevertheless, modulation of the BMP4 induction dose can recapitulate the appropriate patterning. This highlights that robustness to changes in the size of the developing epiblast is attainable, and emphasizes the importance of regulation from other tissues like the vertebrate organizer (the primitive node) in conferring robustness to a developing human embryo. The mechanism that regulates peri-gastrulation-like pattern formation in the geometrically-confined hPSC colonies is one of multiple layers of complexity that would be present in a developing human embryo.

Identity of fate compartments in the peri-gastrulation-like platform

It has been widely reported that BMP4 treatment of hPSCs gives rise to a population that expresses CDX2. However, the identity of this population has been a subject of debate for nearly two decades. Some groups identify this population as trophectoderm(Xu et al. 2002; Warmflash et al. 2014; Li et al. 2013), while other claim that this population is mesodermal in fate (Bernardo et al. 2011; Mendjan et al. 2014; Loh et al. 2016). Interestingly, inhibiting Nodal signaling in the differentiating geometrically-confined colonies results in the abrogation of the primitive-streak-like population and a significant reduction in CDX2 expression. This suggests that at least a subset of the CDX2 positive population within the patterned peri-gastrulation-like fates require Nodal signaling, and is mesodermal in fate.

Additionally, SOX17 expression in differentiating hPSCs has been widely used as a bona fide marker of the endoderm fate(Warmflash et al. 2014; Blauwkamp et al. 2012; Tsakiridis et al. 2014; Green et al. 2011). However, recent studies have shown that SOX17 is important in the induction of the primordial germ cell (PGC) fate in development (Irie et al. 2015; Kobayashi et al. 2017). Although in our report, consistent with the current literature, we define the SOX17 expressing population as endoderm, we have not ruled out the possibility that this population could be the precursor that subsequently gives rise to PGCs.

Of note, the boundary between the ectoderm-like and the primitive-streak-like regions in our platform appears to have cells that express both SOX2, and BRA. We propose two different possibilities that can result in this observation. First, these cells may reflect the population transitioning from a PSC state to a primitive-streak identity which would be upregulating BRA, and downregulating SOX2. Although SOX2 expression in this population would be low, it might still be detectably via immunofluorescence indicating a presence of a SOX2+ BRA+ phenotype, which would stabilize in a SOX2-BRA+ identity. Alternatively, this population might parallel the presumptive neuromesodermal-progenitor-like (NMP-like) population that reside in the node-streak border, caudal lateral epiblast, and the chordoneural hinge sections in the posterior end of the elongating embryo (Gouti et al. 2014; Turner et al. 2014). However, NMPs have been

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proposed to be present relatively later during gastrulation (Henrique et al. 2015), and whether the SOX2-BRA double position population in our peri-gastrulation-like model is, in fact, NMP-like requires further investigation.

The fully defined platform we describe here facilitates investigation of the true identity of all these populations in a developmentally appropriate *in vitro* model system.

Conclusion

In conclusion, we report a defined, *in vitro* model of self-organized human peri-gastrulation-like biological fate patterning, and demonstrate the mechanistic underpinning as a stepwise model of RD, and PI paradigms. This *in vitro* model can be employed to investigate the identity of cell populations that arise out of differentiating hPSCs, especially the populations whose identity has been a subject of debate, while maintaining an appropriate developmental context. We further report that fate acquisition that occurs in a manner consistent with PI can require multiple signaling pathways working in concert. Finally, our data implicates the coordination of multiple tissues to induce morphogen gradients that can scale and maintain robustness to perturbations that may occur in the developing embryo. Consequently, our work not only provides deep insight into one of the earliest stages of human embryonic development, but also into general mechanisms involved in patterning of biological form.

Materials and Methods

Human Pluripotent Stem Cell Culture

The CA1 hPSCs (generously provided by Andras Nagy, Samuel Lunenfeld Research Institute) were cultured on Geltrex (Life Technologies, diluted 1:50) coated 6-well tissue culture plates using mTeSR1™ medium (StemCell Technologies) as per manufacturer's instructions. The cells were passaged at a ratio of 1:12 using ReleSR™ (StemCell Technologies) per manufacturer's instructions. For the first 24h after passage, the cells were cultured in ROCK inhibitor Y-27632 to increase cell viability. The medium was changed every day and passaged every 4 to 5 days or when the cells reached 75-80% confluence.

Preparation of PEG plates to micro-pattern hPSC colonies

Custom sized (110mmx74mm) Nexterion-D Borosilicate thin glass coverslips (SCHOTT) were activated in a plasma cleaner (Herrick Plasma) for 3 minutes at 700 mTorr, and incubated with 1 ml of Poly-L-Lysine grafted Polyethylene Glycol (PLL-g-PEG(5KD), SUSOS,) at a concentration of 1 mg/ml at 37°C overnight. The glass slides were then rinsed with ddH_2O and dried. The desired patterns were transferred to the surface of the PEG-coated side of the coverslip by photo-oxidizing select regions of the substrate using Deep UV exposure for 10 minutes through a Quartz photomask in a UV-Ozone cleaner (Jelight). Bottomless 96-well plates were plasma treated for 3 minutes at 700 mTorr and the patterned slides were glued to the bottomless plates to produce micro-titer plates with patterned cell culture surfaces. Prior to seeding cells onto the plates, the wells were activated with N-(3-Dimethylaminopropyl)-N′-ethylcarbodiimide hydrochloride (Sigma) and N-Hydroxysuccinimide (Sigma) for 20 minutes. The plates were thoroughly washed three times with ddH_2O, and incubated with Geltrex (diluted 1:150) for 4h at room temperature on an orbital shaker. After incubation, the plate was washed with Phosphate Buffered Saline (PBS) at least

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three times to get rid of any passively adsorbed extracellular matrix (ECM) and seeded with cells to develop micro-patterned hPSC colonies.

Cell seeding and induction of peri-gastrulation-like fate patterning

To seed cells onto ECM-immobilized PEG-UV 96-well plates, a single cells suspension of the CA1 line was generated by incubation in 1ml of TryplE (Invitrogen) per well for 3 minutes at 37°C. The TryplE was blocked using in equal volume DMEM + 20% KnockOut Serum Replacement (SR) (Invitrogen) and the cells were dissociated by pipetting to generate a single cell suspension. The cells were centrifuged and re-suspended at a concentration of 1 x 10^6 cells/ml in SR medium supplemented with 20ng/ml bFGF (R&D) and 10μM ROCK inhibitor Y-27632. SR medium consists of 74% DMEM, 1% Penicillin/Streptomycin, 1% non-essential amino acids, 0.1mM β-mercaptoethanol, 1% Glutamax, 2% B27 minus retinoic acid, and 20% SR (all Invitrogen). Wells were seeded in the PEG-patterned 96 well plates at a density of 80,000 cells/well and incubated for 2h at 37°C. After 2h, the medium was changed to SR without ROCKi. An alternative seeding process that has also been tested and provides good results is described in a recent protocol (Deglincerti et al. 2016). When confluent colonies were observed (12-16h after seeding), the peri-gastrulation-like induction was initiated in N2B27 medium supplemented with BMP4 (R&D) (BMP4 dose depended on experimental design). The peri-gastrulation-like pattern formation assay was typically performed with 1000μm diameter colonies 48h following induction, although additional time points and colonies sizes were tested as described in the Results section for specific experiments. N2B27 medium consists of 93% DMEM, 1% Penicillin/Streptomycin, 1% non-essential amino acids, 0.1mM β-mercaptoethanol, 1% Glutamax, 1% N2 Supplement, 2% B27-retinoic acid supplement (all Invitrogen) supplemented with 100ng/ml Nodal (R&D), and 10ng/ml bFGF (R&D).

Single cell data acquisition and analysis of immunofluorescence imaging

The patterned plates were fixed with 3.7% paraformaldehyde for 20 minutes, rinsed three times with PBS and then permeabilized with 100% methanol for 3 minutes. After permeabilization, the patterned colonies were blocked using 10% Fetal Bovine Serum (Invitrogen) in PBS overnight at 4ºC. Primary antibodies were incubated at 4ºC overnight (antibody sources and concentrations shown in Table S1). The following day the primary antibodies were removed, and the plates were washed three times with PBS followed by incubation with the secondary antibodies and DAPI nuclear antibody at room temperature for 1h. Single cell data was acquired by scanning the plates using the Cellomics Arrayscan VTI platform using the TargetActivation.V4 bioassay algorithm. This algorithm utilizes the expression intensity in the DAPI channel to identify individual nuclei in all fields imaged and acquires the associated intensity of proteins of interest localized within the identified region. The single cell data was exported into Context Explorer (CE), a custom software developed in-house for image analysis (Ostblom, *et al*., unpublished). In CE, cell colonies are identified through the DBSCAN algorithm as implemented in Python's Scikit-learn package. Within a colony, each cell is assigned x- and y-coordinates relative to the colony centroid. To create the colony overlay plots, cells from multiple colonies are grouped in hexagonal bins per their relative x- and y-coordinates. These positional bins are color-coded to represent the average protein expression level of all cells within a bin. The color map range is normalized to the lowest and highest expressing hexagonal bins. Spatial expression trends within colonies are also visualized as line plots, where cells are grouped by the Euclidean distance between a cell and the centroid of the colony. For each colony, the average expression value of all cells within a distance bin is computed. The line plots describe the mean expression value, standard deviation and 95% confidence interval (CI) between colonies as indicated in figure legends. The line plots depicting radial trends of proteins of interest in individual colonies were acquired through the sections of the colonies depicted (shown as white rectangles in the figures) in Fiji (ImageJ). The plot profile extracted was then run through a Savitzky-Golay smoothing filter in Matlab and represented as a function of radial distance.

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The quantification of RD-like patterns in differentiating, geometrically-confined hPSC colonies of 3mm diameters was performed using SIESTA, an image analysis platform (Fernandez-Gonzalez & Zallen 2011; Leung & Fernandez-Gonzalez 2015), and custom scripts written in MATLAB (Mathworks) using the DIPImage toolbox (TU Delft, The Netherlands). To quantify the periodicity of free BMP4 ligands in the computer model, or pSMAD1 and BRA signals in colonies, we used the Fourier transform. To avoid quantifying background noise, we thresholded the BRA and pSMAD1 signals, and the pixels with intensity lower than the mean of the image plus one standard deviation were set to zero. We measured the signal intensity in the colonies along radii separated by an angle of 30°. The signals were detrended by subtracting the mean pixel value, and smoothened using a Gaussian of $\sigma = 3$ pixels for the colonies and $\sigma = 20$ for the model, consistent with the different spatial resolution of experimental and model images. The edge of the BRA colonies was excluded from the analysis to avoid artefacts generated by accumulation of migrating cells. Finally, we used the Fourier transform to decompose each signal into its constituent frequencies and computed a characteristic period, defined as the inverse of the dominant frequency. To be able to compare periods between colonies and the model, the length of each radius was normalized to 1. Periods greater than one indicated lack of periodicity.

Quantitative PCR analysis

RNA extraction was performed using Qiagen RNAeasy miniprep columns according to the manufacturer's protocol, and the cDNA was generated using Superscript III reverse transcriptase (Invitrogen) according to the manufacturer's instructions. The generated cDNA was mixed with primers for the genes of interest and SYBR green mix (Roche, Sigma) and the samples were run on an Applied Biosystems QuantStudio 6 flex real-time PCR machine. Relative expression of described genes was determined by the delta–delta cycle threshold (ΔΔCt) method with the expression of GAPDH as an internal reference. Primer sequences used are provided in Supplementary Information (Table S2).

Statistics and data analysis

All gene expression results were expressed as mean (+S.D.). Statistical tests for gene expression results were performed using two-tailed Student's t-test assuming unequal variance between datasets. The statistical significance for fate acquisition results were calculated either using one-way ANOVA (Dunnett's post-hoc test) or the Kruskal-Wallis test as described in the legends. The calculations for Student's t-tests were performed in Excel. The one-way ANOVA, and the Kruskal-Wallis test calculations were performed in Prism. To evaluate sample means of periodic distributions predicted by the RD-model with the distributions experimentally observed, we used a non-parametric Mann–Whitney U test; and we used the Kolmogorov–Smirnov test was used to compare sample distributions.

siRNA transfection protocol

All siRNA transfection was performed with CA1 hPSCs seeded on either PEG plates as described above, or 24 well plates for the qPCR control experiments to validate the siRNA specificity. Noggin specific siRNAs – NOG Silencer (Thermo Fisher), and siRNA NOG (Santa Cruz Biotechnology), mixed at 1:1 ratio, at a total concentration of 40nM; or scrambled control siRNA (Santa Cruz Biotechnology) at a concentration of 40nM. Transfection was performed using EditPro™ Stem Transfection Reagent (MTI-Globalstem). Specifically, 1µl (per 24 well) or 0.2µl (per 96 well) EditPro™ Stem Transfection Reagent was diluted with 50µl or 10µl Opti-MEM I reduced serum medium (Thermo Fischer) respectively. Diluted reagent was incubated with siRNA for 15 minutes at RT and subsequently added to cells containing 500µl (24 well) and 100µl (96 well) culture medium without Penicillin/Streptomycin. Medium was replaced with N2B27+BMP4 18 hours after

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transfection. Cells were analyzed 24 hours after transfection by qPCR (24 well) for NOGGIN gene expression, or by microscopy (96 well) for pSMAD1 spatial trends.

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Figures

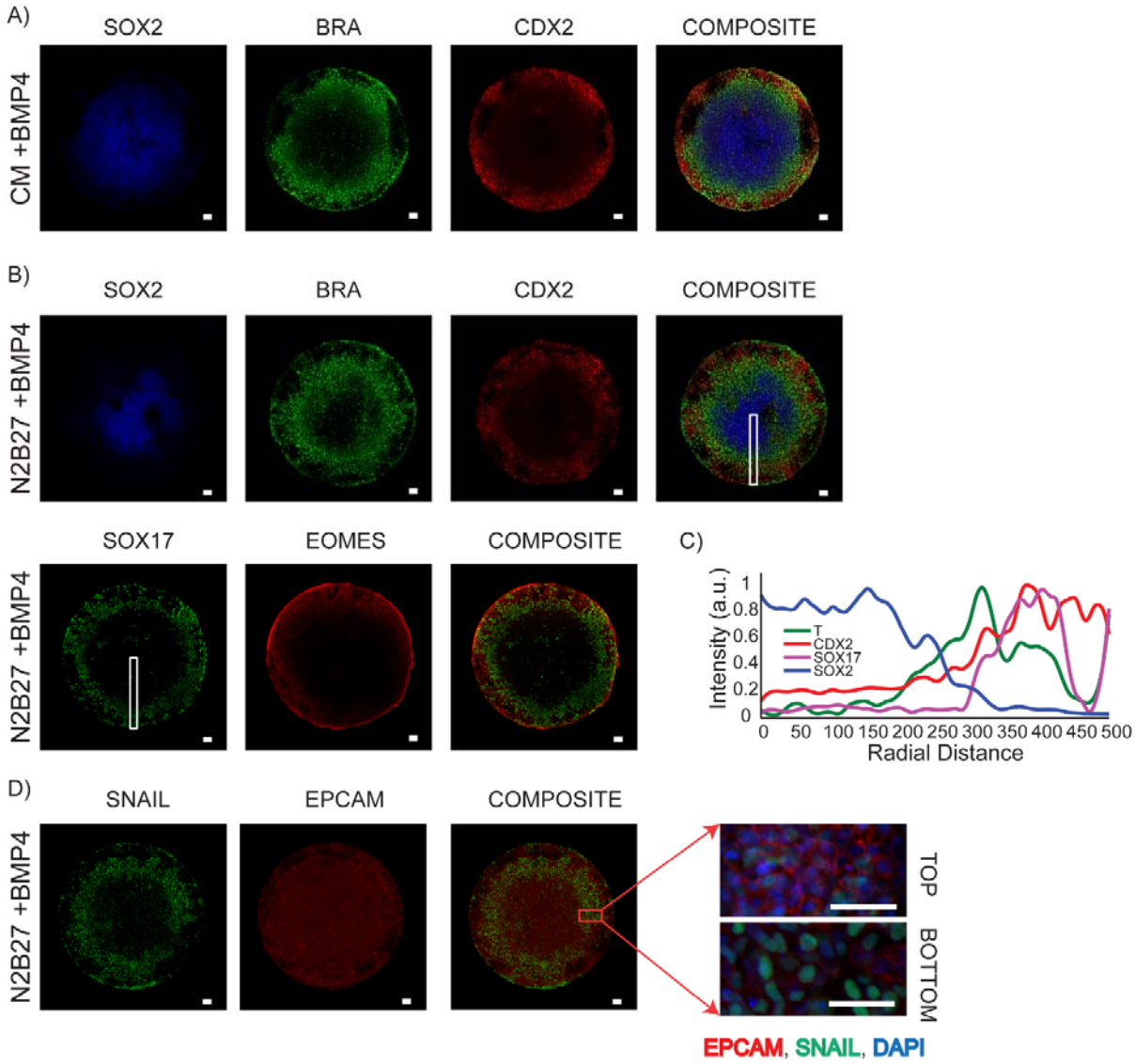
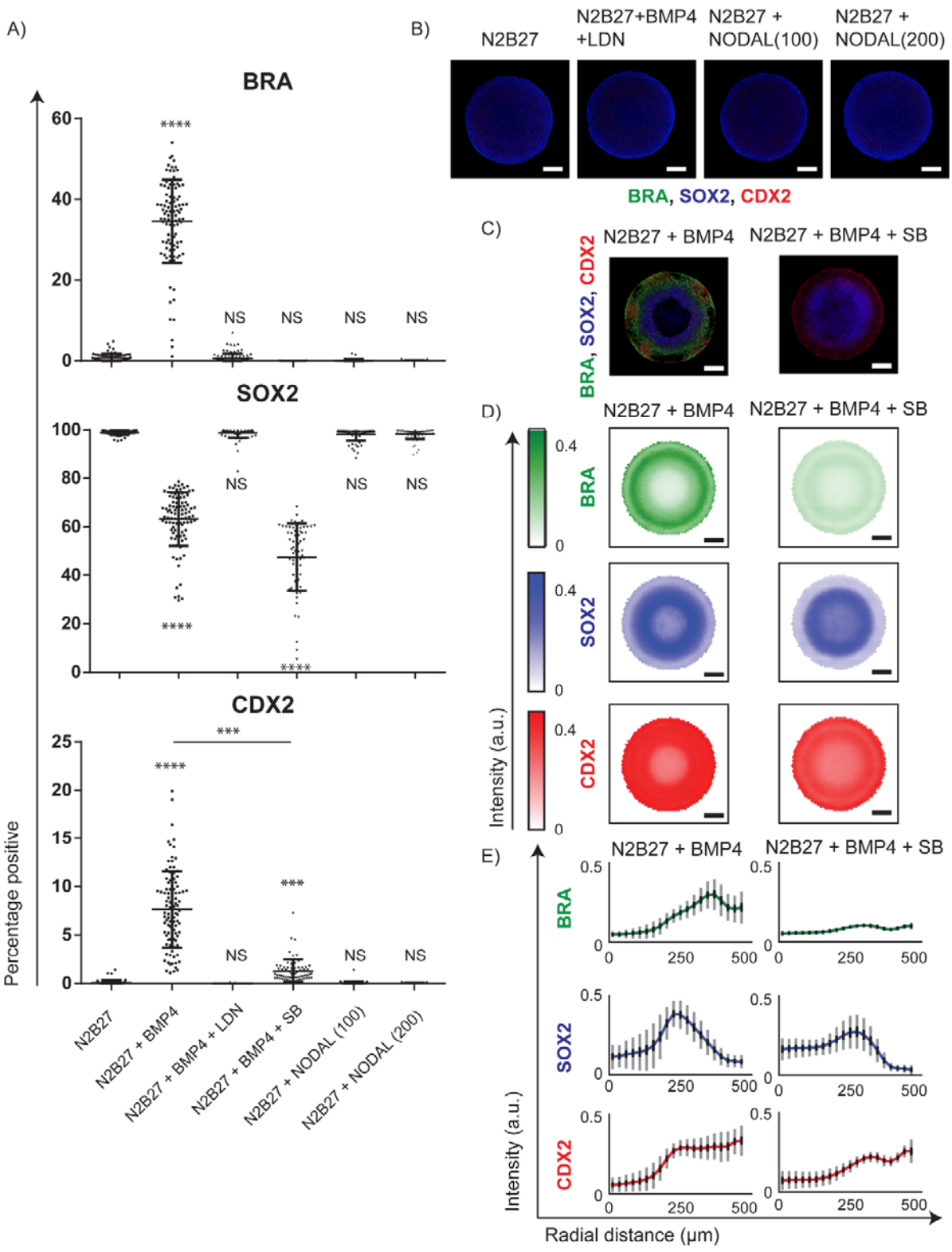


Figure 1: **Defined peri-gastrulation-like patterning induction in differentiating hPSC colonies**. Representative immunofluorescence images of: A) fate patterning of SOX2, BRA, and CDX2 in BMP4 supplemented CM as previously reported (Warmflash et al. 2014; Etoc et al. 2016); B) fate patterning in BMP4 supplemented N2B27 medium stained for SOX2, BRA, and CDX2, SOX17, and EOMES. C) Spatial trends for intensity of expression of SOX2, BRA, CDX2, and SOX17 in regions marked by white rectangle in B and C (Average trends of replicates shown in Fig S1, S2). E) Representative image of SNAIL and EPCAM staining in micro-patterned colony differentiated in N2B27 shows that mesenchymal

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marker expressing cells in the primitive streak region are located underneath an epithelial layer. Scale bars represent 50μm.

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Figure 2**: Nodal signaling is required for primitive streak specification, but does not induce differentiation and fate patterning in geometrically-confined hPSC colonies**. A) Percentage of cells expressing BRA, SOX2, and CDX2 in N2B27, which includes 100ng/ml of NODAL (n=73), N2B27+BMP (n=100), N2B27+BMP+LDN (n=101), N2B27+BMP+SB (n=67), N2B27+100ng/ml Nodal (n=68), and N2B27+ 200ng/ml Nodal (n=65). The experiment was performed twice. **** ($p<0.0001$), NS ($p>0.05$). The p values were calculated using one-way ANOVA (Dunnett's post-hoc test). B) Representative images of colonies cultured in N2B27, BMP+LDN, Nodal (100), and Nodal (200) stained for BRA, SOX2, and CDX2. C) Representative immunofluorescent images of colonies cultured in BMP4, and BMP4+SB stained for BRA, SOX2, and CDX2. D) SOX2, BRA and CDX2 expression averages for 100 colonies cultured in BMP4, and 67 colonies cultured in BMP4+SB. E) Average radial trends from D) for BMP4 and BMP4+SB conditions. Standard deviations shown in grey, and 95% confidence intervals in black. Scale bars represent 200μm.

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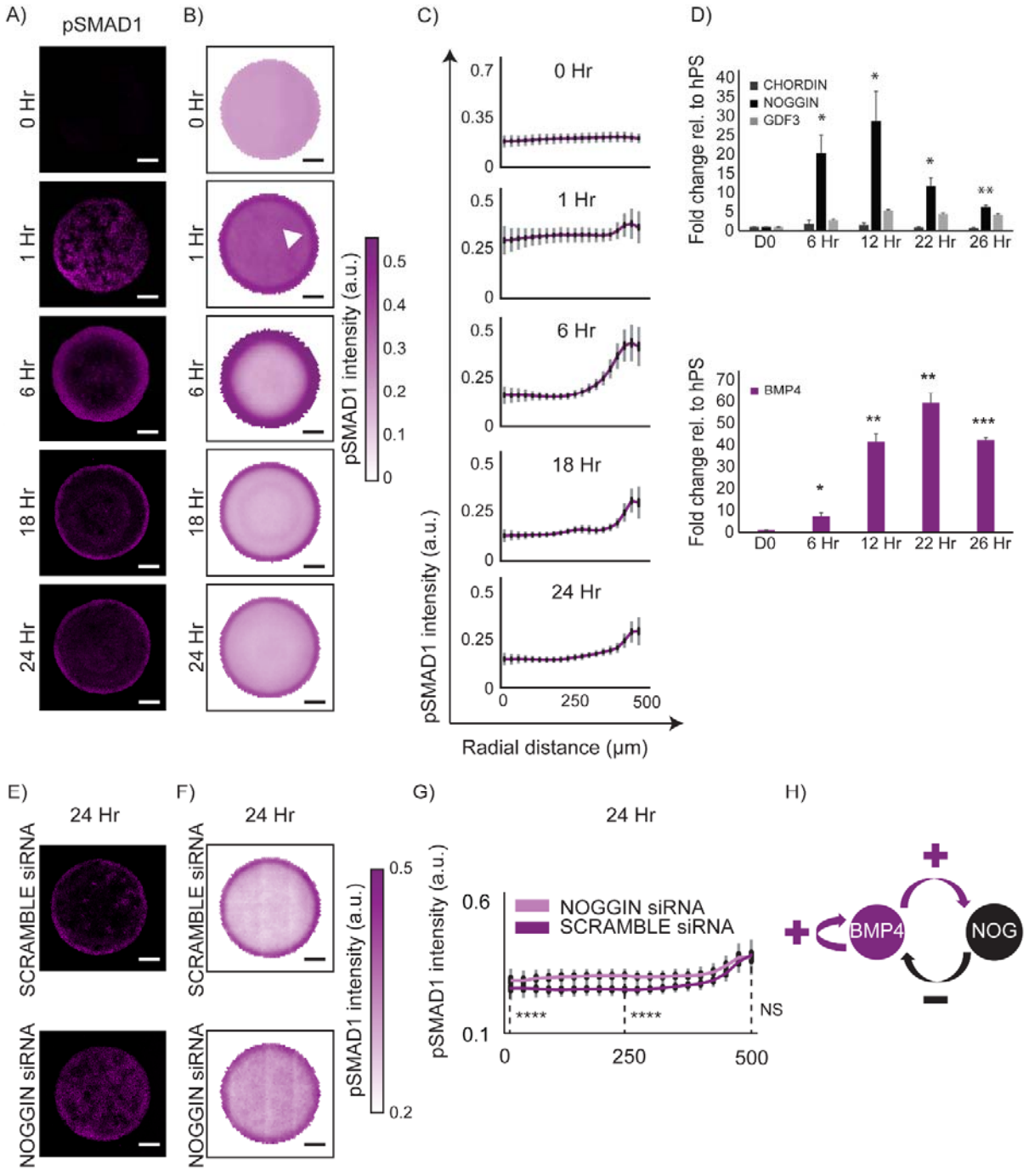


Figure 3: **pSMAD1 gradient self-organization in differentiating colonies suggests the presence of a BMP4-NOGGIN RD network**. A) Representative immunofluorescence images of colonies stained for pSMAD1 at 0, 1, 6, 18, and 24h after BMP induction. B) Average pSMAD1 intensity represented as overlays for 202, 193, 105, 100, 105 colonies for the respective induction times. Data were collected from two experiments. White arrowhead marks the region of relatively lower average pSMAD1 activation. C) The

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average radial trends of pSMAD1 activity at each induction duration. Standard deviations are shown in grey, and 95% confidence intervals in black. D) Temporal gene expression profiles for BMP signaling inhibitors (CHORDIN, NOGGIN, and GDF3) and BMP4 at 6, 12, 22, and 26h. Data shown as mean and standard deviation (S.D.) of three independent experiments. * $p<0.05$, ** $p<0.01$, *** $p<0.001$. E) Representative pSMAD1 immunofluorescence images for colonies treated with SCRAMBLE siRNA and NOGGIN siRNA after 24h of BMP4 treatment. F) Average pSMAD1 expression in colonies treated with SCRAMBLE siRNA, NOGGIN siRNA (64, and 62 colonies respectively). Data were collected from two experiments. G) Radial trends of pSMAD1 signaling distribution from (F) for SCRAMBLE, and NOGGIN siRNA conditions. Standard deviations are shown in grey, and 95% confidence intervals in black. **** $p<0.0001$, NS ($p>0.05$). The p values were calculated using Mann-Whitney U test. H) Overview of the implicated BMP4-NOGGIN RD network. Scale bars represent 200μm.

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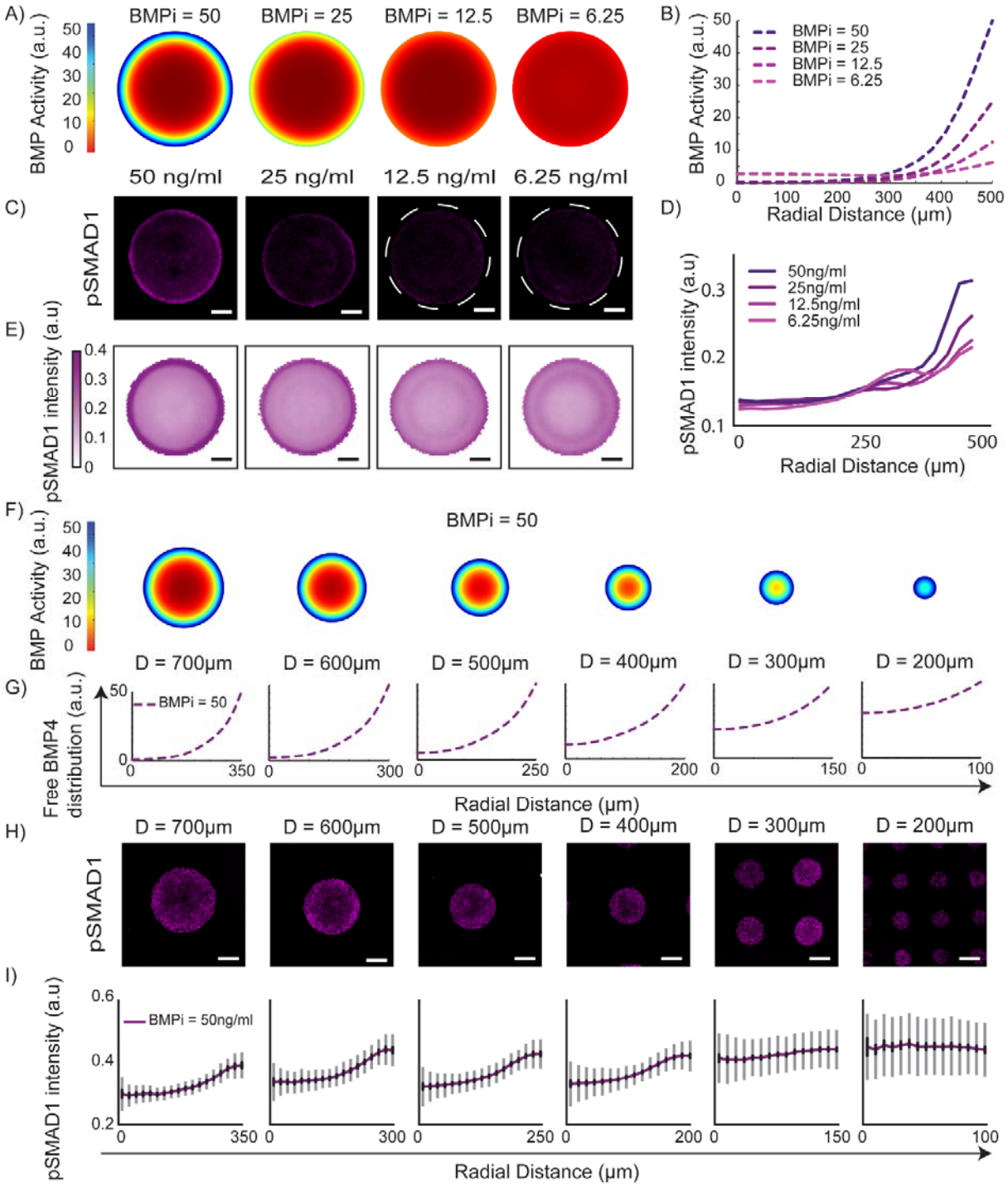


Figure 4: **BMP4-NOGGIN RD model predicts pSMAD1 gradient response to colony size and BMP4 dose perturbations**. A) Predicted distribution of free BMP4 ligands in colonies of 1000μm diameter as a function of varying BMPi concentration (50ng/ml, 25ng/ml, 12.5ng/ml, and 6.25ng/ml). B) Line plots of predicted distributions. C) Representative immunofluorescence images of micro-patterned colonies stained

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for pSMAD1 24h after induction with varying BMPi conditions (50ng/ml, 25ng/ml, 12.5ng/ml, and 6.25ng/ml). D) pSMAD1 signaling distribution represented as a function of colony radius and E) averages of 143,119, 129, 163 colonies for respective conditions. Data were pooled from two experiments. F) Predicted distribution of free BMP4 ligands, following induction with 50ng/ml BMP4, as a function of colony diameter (700μm, 600μm, 500μm, 400μm, 300μm, and 200μm). G) Graphical depiction of predicted distributions in F. H) Representative immunofluorescence images of colonies stained for pSMAD1 24h after induction with 50ng/ml of BMP4. I) Average pSMAD1 expression of 87, 118, 178, 261, 437, and 373 colonies for respective conditions. J) pSMAD1 signaling distribution represented as a function of the colony radius. Data were collected from two experiments. Scale bars represent 200μm.

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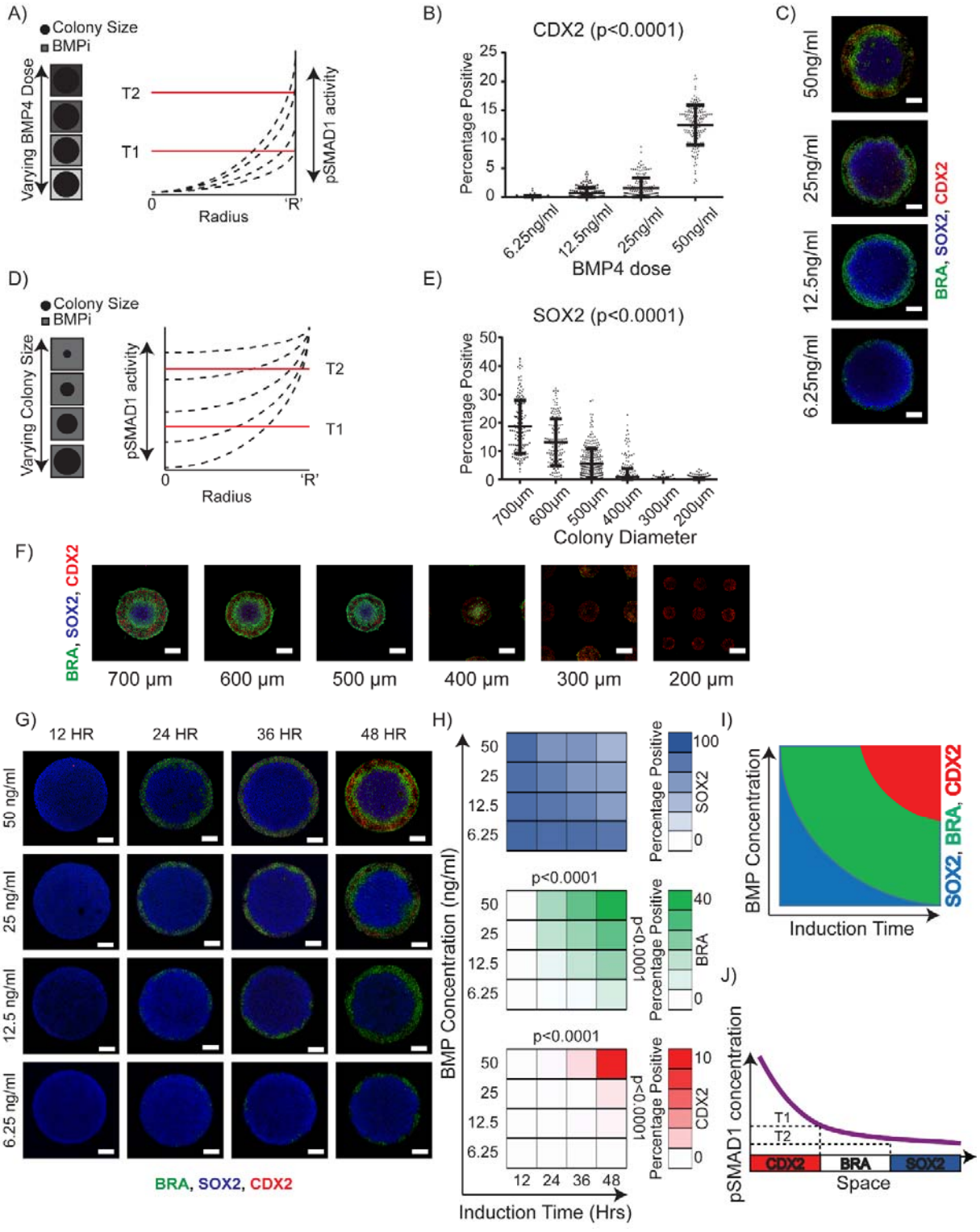
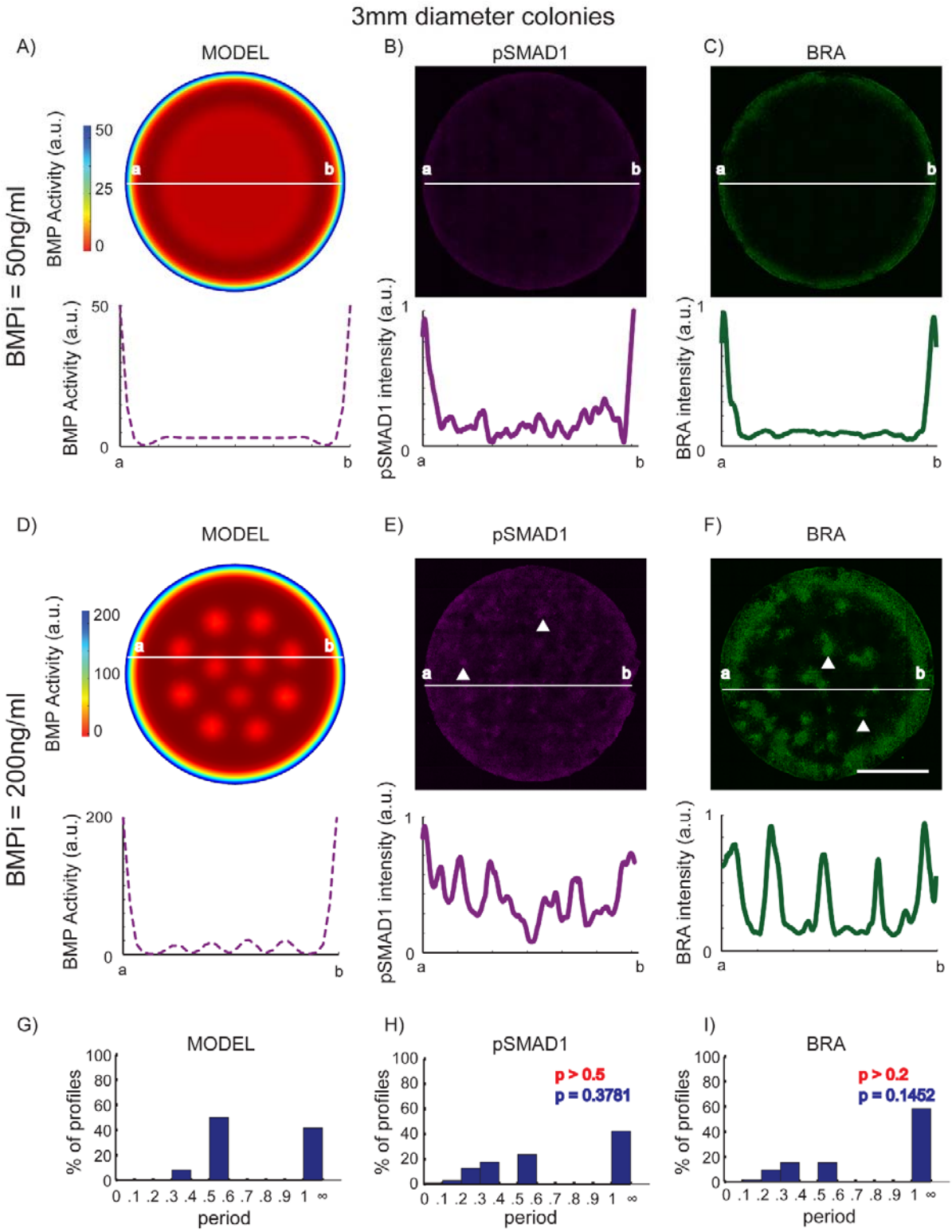


Figure 5: **Fate patterning in hPSC colonies arises in a pSMAD1 threshold dependent manner**. A) Overview of the experimental setup: varying BMPi while maintaining colony size constant perturbs pSMAD1 signaling at the colony periphery. T1 and T2 represent putative thresholds that determine fate patterning.

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B) Quantification of cells expressing CDX2 in 1000μm colonies induced to differentiate at varying BMPi concentrations (6.25ng/ml, 12.5ng/ml, 25ng/ml, and 50ng/ml) (p value calculated using the Kruskal Wallis test. Number of colonies are 136, 168, 169, and 156 for the respective conditions. Results pooled from two separate experiments. C) Representative immunofluorescence images of SOX2, BRA, and CDX2 expression in geometrically confined 1000μm diameter colonies differentiated in 6.25ng/ml, 12.5ng/ml, 25ng/ml and 50ng/ml of BMP4. D) Overview of the experimental setup: colony size is varied while maintaining constant BMPi perturbs the level of pSMAD1 signaling in the colony center. T1 and T2 represent putative thresholds that determine fate patterning. E) Quantification of cells expressing SOX2 in colonies of varying diameters (700μm, 600μm, 500μm, 400μm, 300μm, 200μm) differentiated in BMPi = 50ng/ml (p value calculated using the Kruskal Wallis test). Number of colonies analyzed from two separate experiments were 144, 160, 279, 466, 789, and 1607 for the respective conditions. F) Representative immunofluorescent images of SOX2, BRA, and CDX2 expression in geometrically confined colonies of varying diameters (700μm, 600μm, 500μm, 400μm, 300μm, and 200μm). G) Representative immunofluorescence images of SOX2, BRA, and CDX2 stained 1000μm diameter colonies differentiated at varying BMPi concentrations (50ng/ml, 25ng/ml, 12.5ng/ml, and 6.25ng/ml) and induction times (12, 24, 36, and 48h). H) Average percentage of cells expressing SOX2, BRA, and CDX2 in hPSC colonies. Each condition had over 140 colonies. Data were pooled from two experiments. The p values were calculated for the concentration of 50ng/ml and induction time of 48h using the Kruskal Wallis test. I) Overview of peri-gastrulation-like fate acquisition arising as a function of both morphogen concentration and induction time. J) Representation of fate acquisition arising in a manner consistent with the cardinal Positional Information paradigm. Scale bars represent 200μm.

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Figure 6: **High BMP4 dose in induction media recapitulates stereotypic RD-like periodic patterns in 3mm diameter hPSC colonies**. A) Model predictions for spatial profile of BMP activity for colonies of 3mm diameter at BMPi = 50ng/ml. Experimental data showing representative immunofluorescent images and spatial expression profiles for B) pSMAD1, and C) BRA. D) Model predictions for spatial profile of BMP activity for 3mm diameter colonies differentiated at a higher dose of BMPi (200ng/ml). Experimental data showing representative immunofluorescent images and spatial expression profiles for E) pSMAD1, and F) BRA. Histogram of dominant periods identified in the G) computational model of RD-like BMP pattern formation, and experimentally identified periods for H) pSMAD1 (n=28 colonies pooled from 3 experiments), and I) BRA (n = 47 colonies pooled from 3 experiments) expression. The p-values in red were calculated using Mann-Whitney U test, and in blue were calculated using Kolmogorov-Smirnov test. Periods greater than 1 indicate no periodicity. Scale bars represent 1mm.

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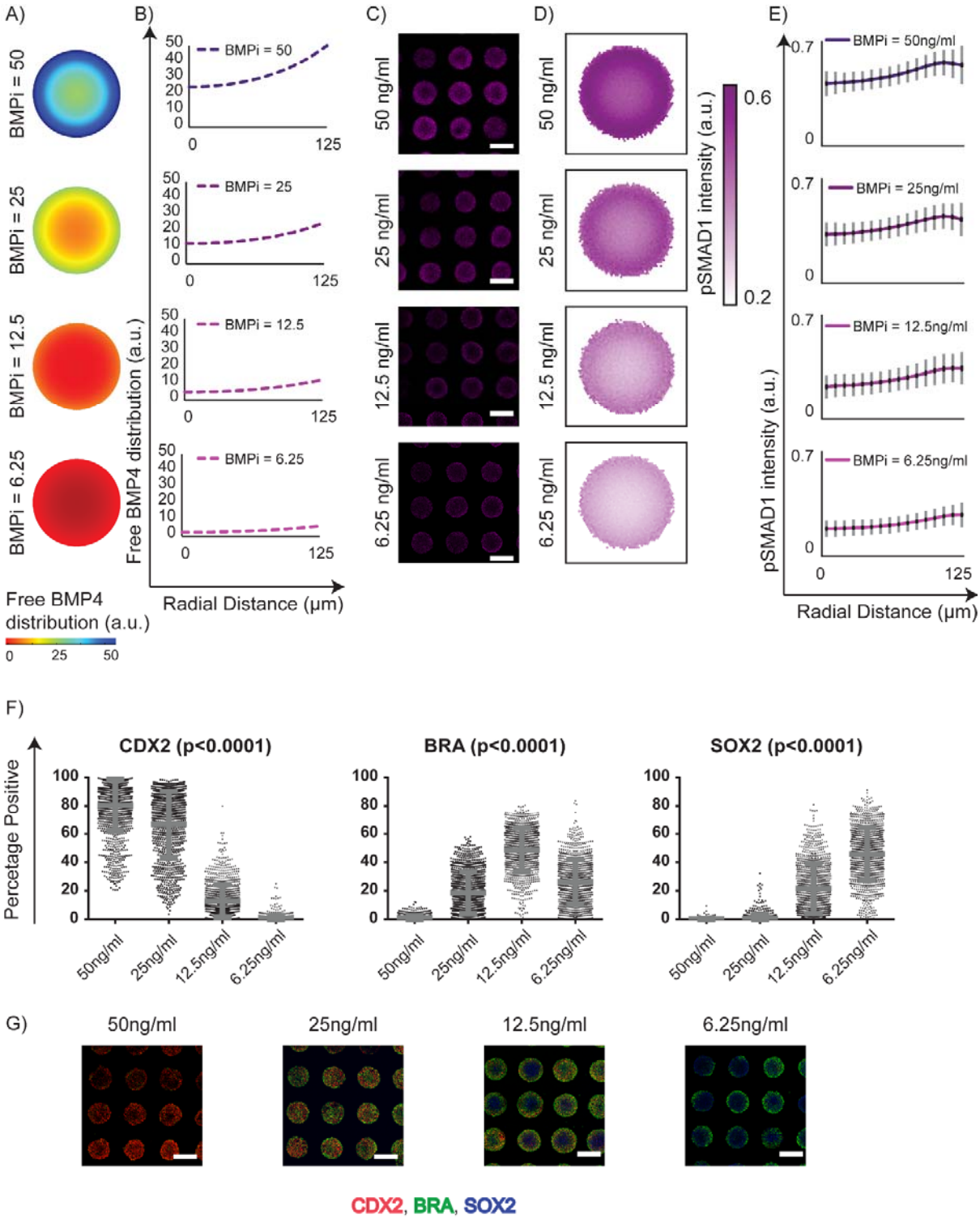


Figure 7: **Low BMP4 dose in induction media rescues fate patterning in 250μm diameter colonies**. A) Model predictions for gradient formation of free BMP4 ligands distribution in 250μm diameter colonies in response to varying BMPi doses. B) Line plot representation of free BMP4 ligands as a function of the colony radius. C) Representative immunofluorescence images of colonies stained for pSMAD1 24h after

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induction with varying doses of BMP4. D) Average pSMAD1 expression for 976, 638, 475, and 689 colonies for respective conditions. E) Signaling distribution represented as a function of the colony radius. The results were collected from two experiments. F) Quantification of percentage of cells in each colony expressing CDX2, BRA, and SOX2 in 250μm diameter colonies induced to differentiate at varying BMPi concentrations (6.25ng/ml, 12.5ng/ml, 25ng/ml, and 50ng/ml), p-value calculated using the Kruskal-Wallis test. Number of colonies are 1024, 1102 1134, and 1135 for the respective conditions. Results pooled from two separate experiments. G) Representative immunofluorescence images of SOX2, BRA, and CDX2 expression in geometrically confined 250μm diameter colonies differentiated in 6.25ng/ml, 12.5ng/ml, 25ng/ml and 50ng/ml of BMP4. Scale bars represent 200μm.

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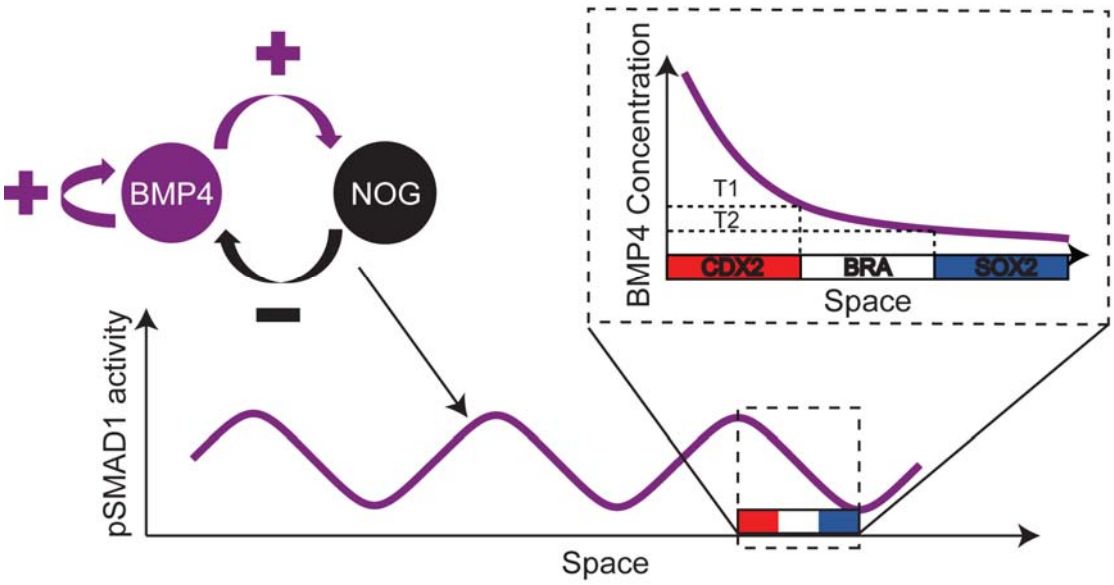
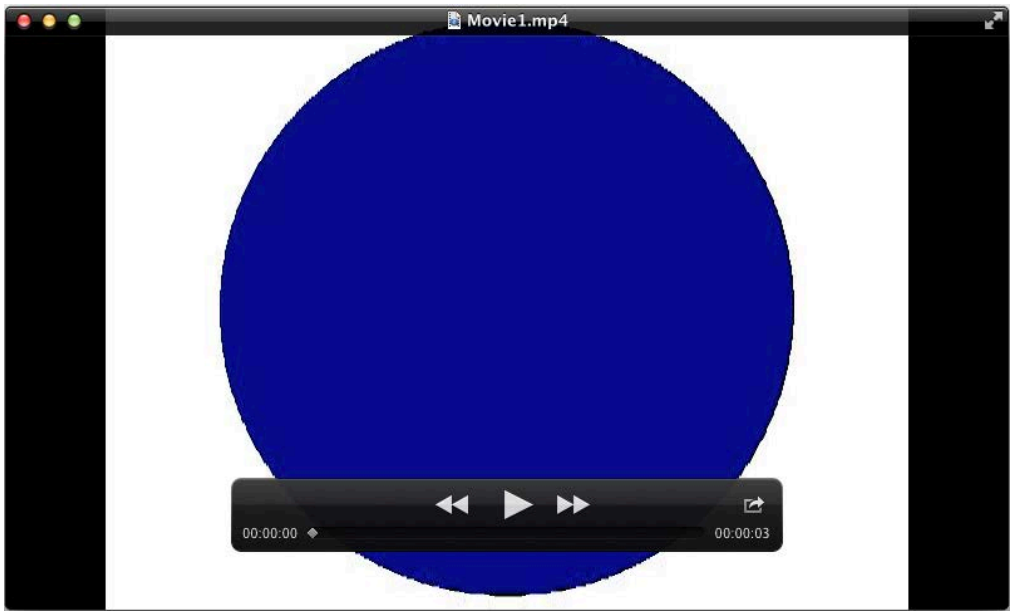


Figure 8: **Mechanism of peri-gastrulation-like fate patterning in geometrically confined hPSC colonies**. A BMP4-NOGGIN RD network induces a radial and periodic pSMAD1 activity gradient in geometrically confined hPSC colonies. The fate acquisition due to the pSMAD1 gradient follows the classical PI paradigm.

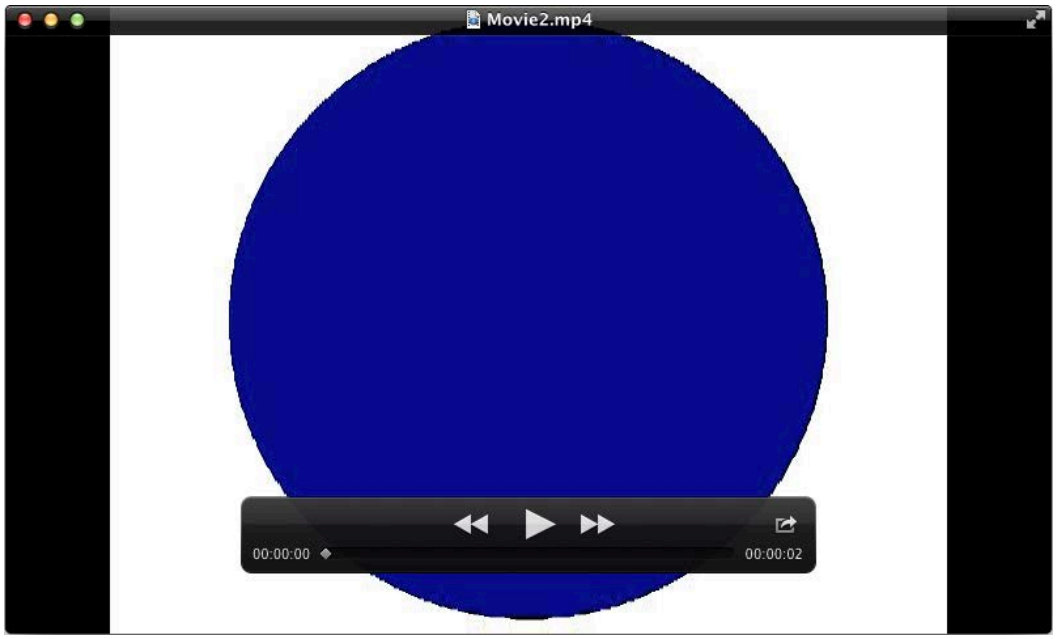
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Movies



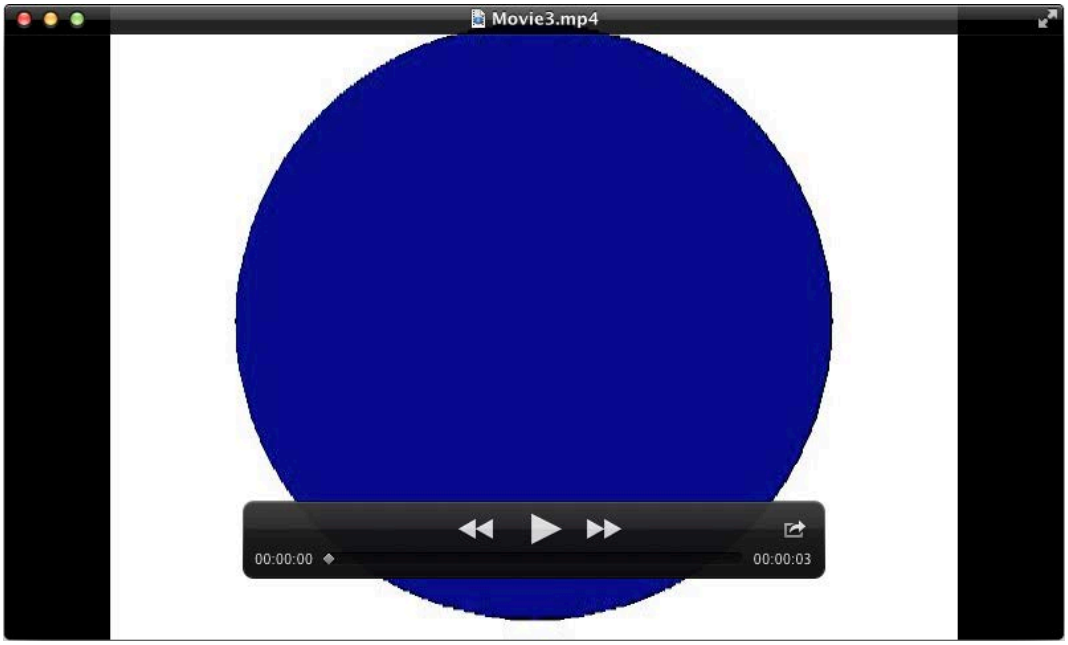
Movie 1: Reaction-Diffusion model of free BMP4 distribution in differentiating hPSC colonies. The video shows the evolution of free BMP4 distribution in differentiating hPSC colonies in accordance with the BMP4-NOGGIN RD model in a differentiating colony of a 1000µm diameter.

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Movie 2: Reaction-Diffusion model of predicts BMPi = 50ng/ml would not generate periodic patterns of free BMP4 distribution in differentiating hPSC colonies of 3mm in diameter. The video shows the evolution of free BMP4 distribution in accordance with the BMP4-NOGGIN reaction diffusion model in differentiating hPSC colonies of 3mm in diameter when induced to differentiate with BMPi = 50ng/ml.

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Movie 3: Reaction-Diffusion model of predicts induction of periodic waves of free BMP4 ligands distribution when hPSC colonies of 3mm in diameter are differentiated in BMPi = 200ng/ml. The video shows the evolution of free BMP4 distribution in accordance with the BMP4-NOGGIN reaction diffusion model in differentiating hPSC colonies of 3mm in diameter when induced to differentiate with BMPi = 200ng/ml.

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Supplementary Tables

Table S1: Antibodies used for immunofluorescence experiments

Antibody Target	Company and catalog ID	Dilution
CDX2	Abcam (ab15258), Cedarlane (MU392A-UC)	Abcam (1:50), Cedarlane (1:400)
BRA	R&D (AF2085)	1:500
SOX2	Cell Signaling (3579S), R&D (MAB2018)	Cell Signaling (1:200), R&D (1:500)
SOX17	R&D (AF1924)	1:500
EOMES	Abcam (ab23345)	1:500
EPCAM	R&D (SC026) Kit	1:10
SNAIL	R&D (SC026) Kit	1:10
pSMAD1	Cell Signaling (9516S)	1:200
NANOG	Cell Signaling (4903S)	1:200

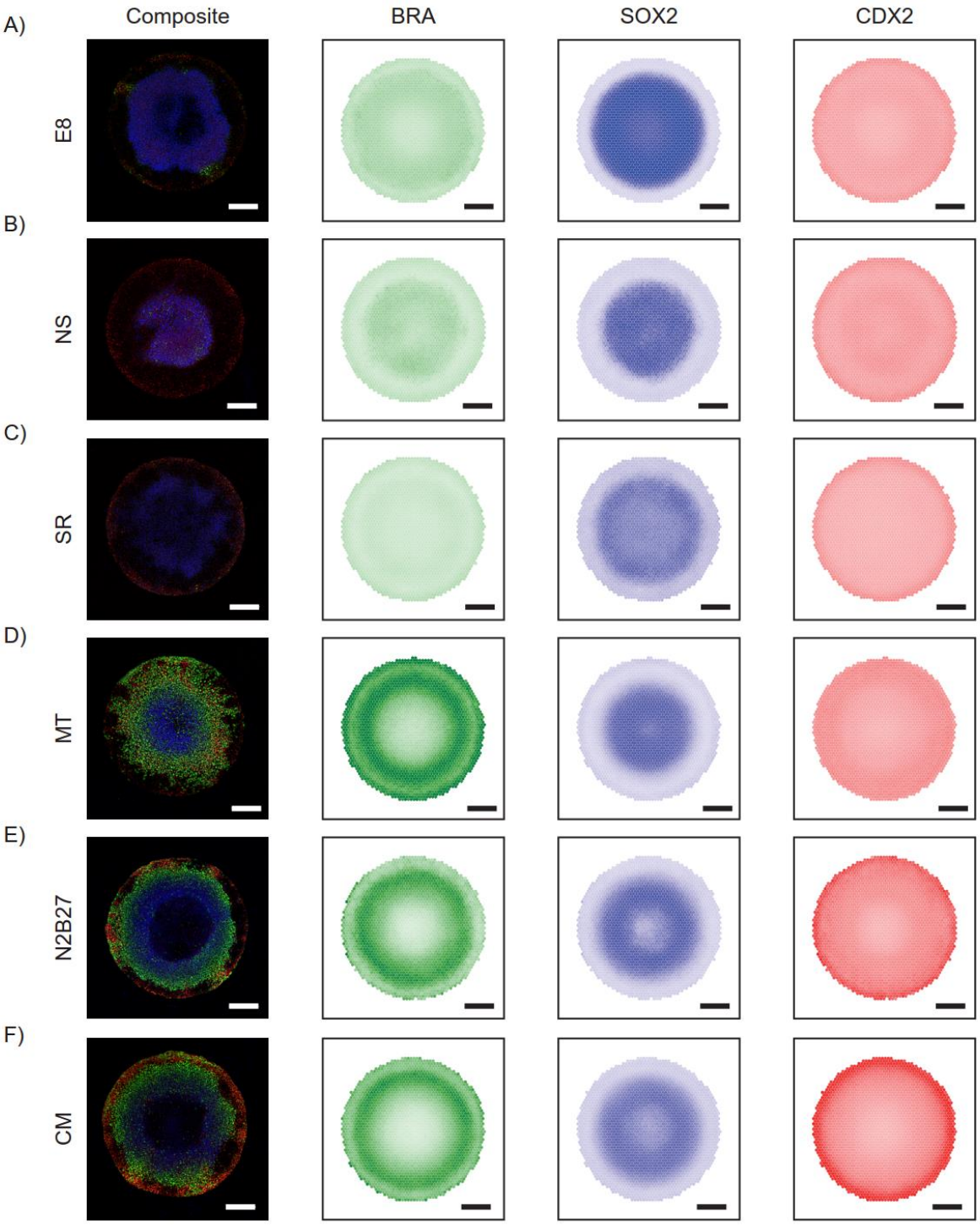
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Table S2: Primers used in qPCR experiments

Primer target	Sequence or catalog ID
BMP4 (fwd)	ATGATTCCTGGTAACCGAATGC
BMP4 (rev)	CCCCGTCTCAGGTATCAAACT
NOGGIN (fwd)	NM_005450.4 (GeneCopoeia Cat # HQP054071)
NOGGIN (rev)	NM_005450.4 (GeneCopoeia Cat # HQP054071)
CHORDIN (fwd)	NM_001304473.1 (GeneCopoeia Cat # HQP067561)
CHORDIN (rev)	NM_001304473.1 (GeneCopoeia Cat # HQP067561)
FOLLISTATIN (fwd)	NM_013409.1 (GeneCopoeia Cat # HQP000565)
FOLLISTATIN (rev)	NM_013409.1 (GeneCopoeia Cat # HQP000565)
GDF3 (fwd)	GTACTTCGCTTTCTCCCAGAC
GDF3 (rev)	GCCAATGTCAACTGTTCCCTT
CERL (fwd)	CTTCTCAGGGGGTCATCTTG
CERL (rev)	TCCCAAAGCAAAGGTTGTTC

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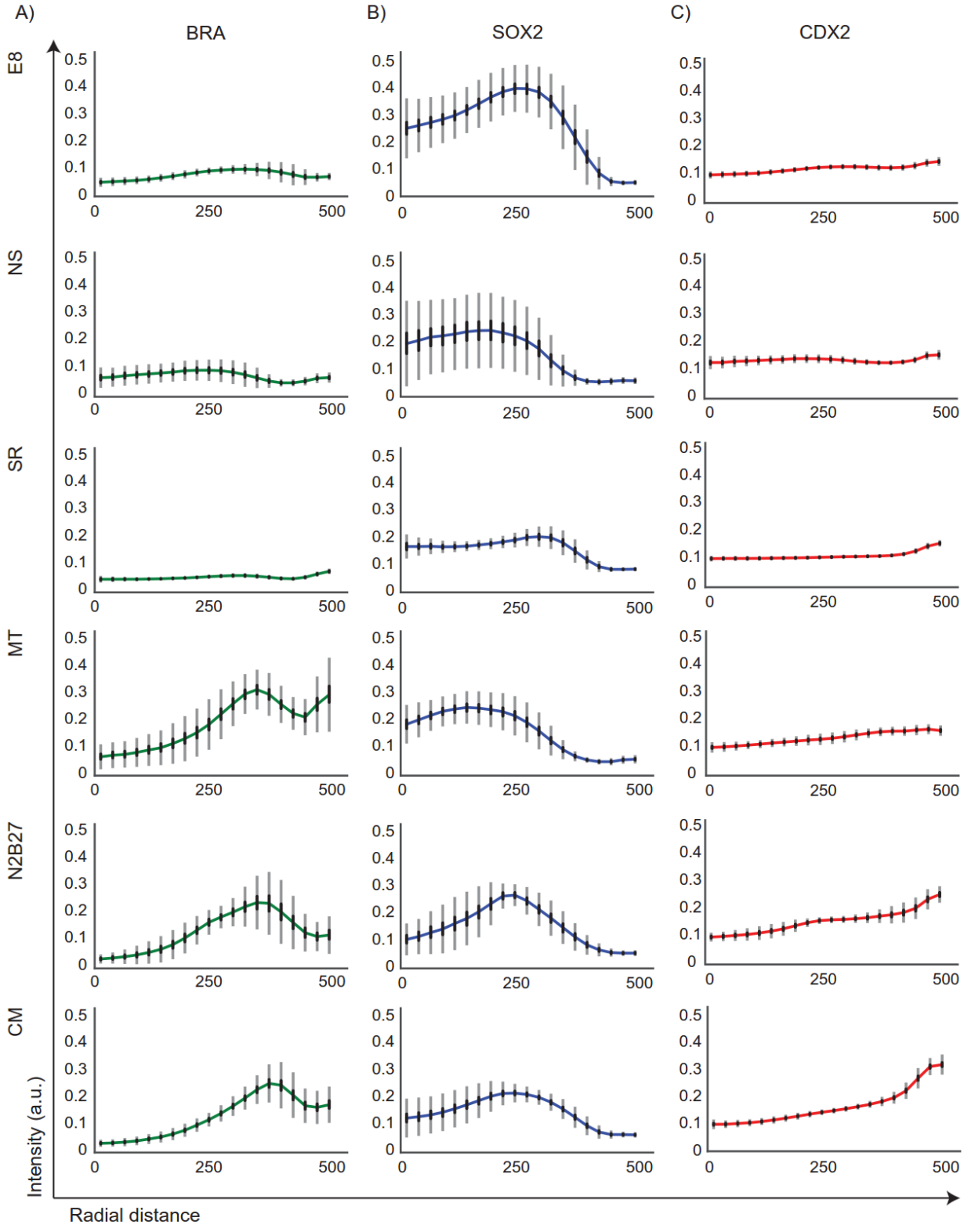
Supplementary Figures



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Figure S1: Peri-gastrulation-like fate patterning in multiple basal medium conditions. Representative composite images and spatial expression average for SOX2, BRA and CDX2 staining in geometrically-confined hPSC colonies differentiated in BMP4 supplemented A) E8, B) Nutristem (NS), C) SR, D) mTeSR, E) N2B27, and F) CM. Scale bars represent 200μm.

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Figure S2: Spatial trends of CDX2, BRA, and SOX2 observed in different basal media. Radial trends of A) BRA, B) SOX2, and C) CDX2 in BMP4 supplemented E8, NS, SR, MT, N2B27, and CM. Standard deviations shown in grey, and 95% confidence intervals shown in black.

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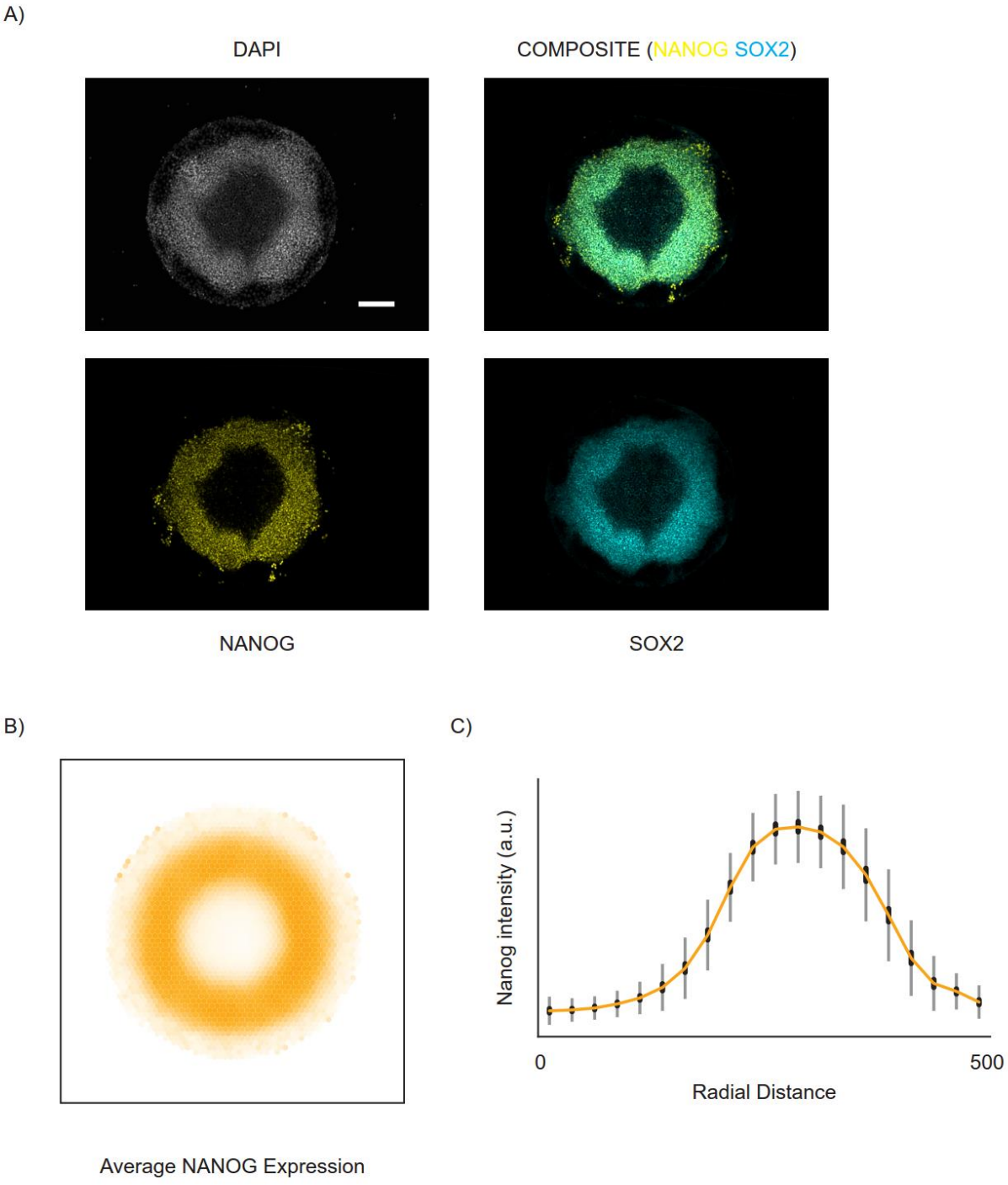


Figure S3: Nanog does not co-localize with SOX2 expression at the center of differentiating hPSC colonies. A) Representative immunofluorescent images of colonies stained for DAPI, SOX2, and NANOG of geometrically confined hPSC colonies cultured in BMP4

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supplemented N2B27. NANOG expression of 95 colonies shown as B) Average map, and C) line-plots of the average radial trend. Standard deviations shown in grey, and 95% confidence intervals shown in black. Data pooled from two experiments. Scale bars represent 200μm.

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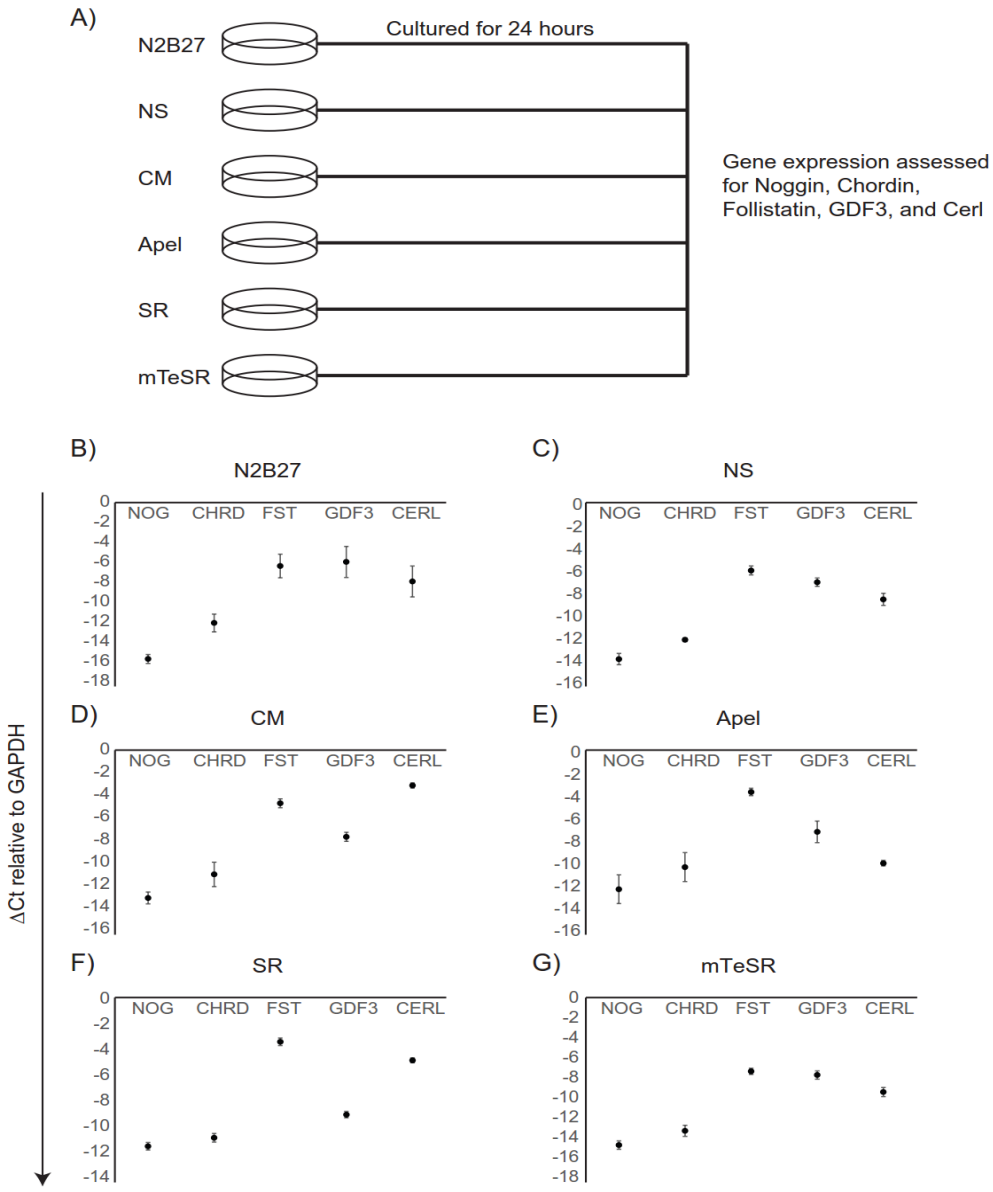


Figure S4: Basal expression of BMP inhibitors in hPSCs during routine culture. A) Overview of experimental setup – hPSCs were cultured in a variety of media conditions for 24 hours and gene expression was assessed for candidate inhibitors of BMP signaling. Expression of various inhibitors (NOG – NOGGIN, CHRD – CHORDIN, FST – FOLLISTATIN, GDF3, and CERL – CERBERUS-Like) of BMP signaling under basal conditions shown as ΔCt relative to GAPDH for CA1 cells cultured in B) N2B27, C) Nutristem (NS), D) Conditioned Medium (CM), E) Apel, F) Serum Replacement medium (SR), and G) mTeSR for 24 hours. Data are shown as mean (S.D) for three independent experiments

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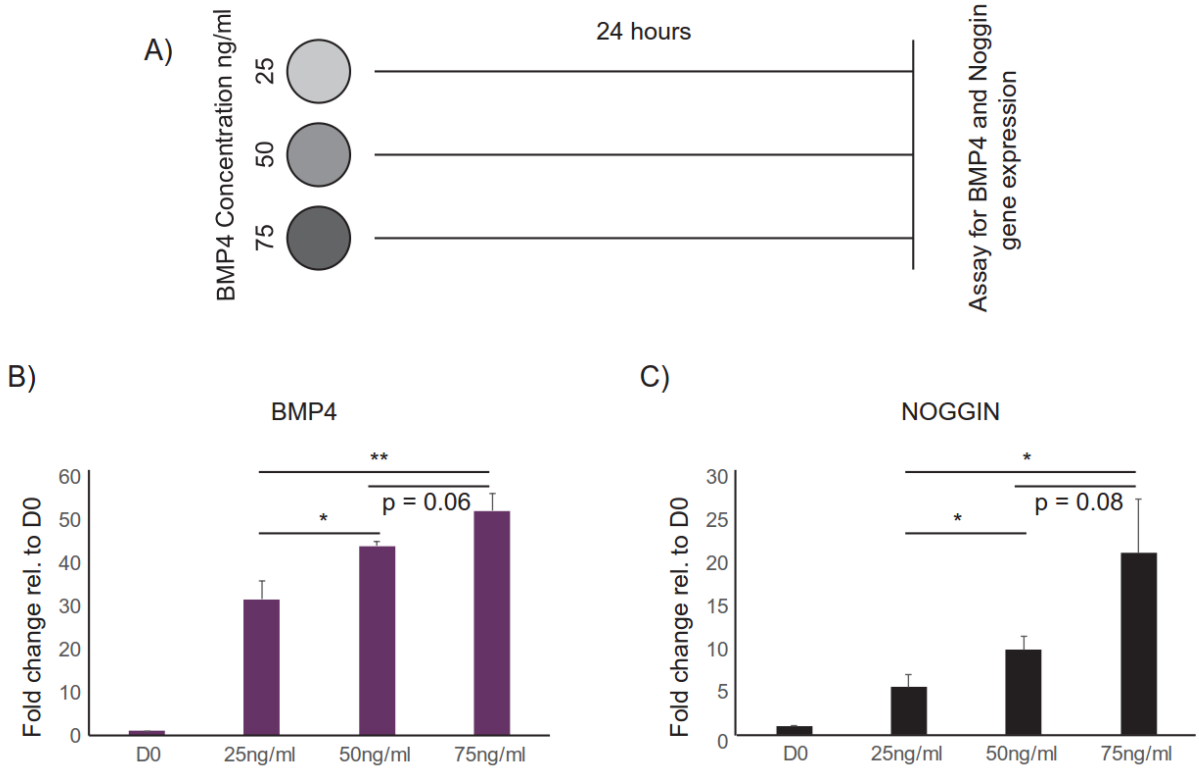


Figure S5: BMP4 and NOGGIN upregulation occur in a BMP4 induction dose-dependent manner. A) Experimental overview: Gene expression data gathered at 24 hours following induction at varying concentrations of BMP4. B) BMP4-induced expression of BMP4. C) BMP4 induced expression of NOGGIN. Data represented as mean and S.D. of three biological replicates. * $p<0.05$, ** $p<0.01$.

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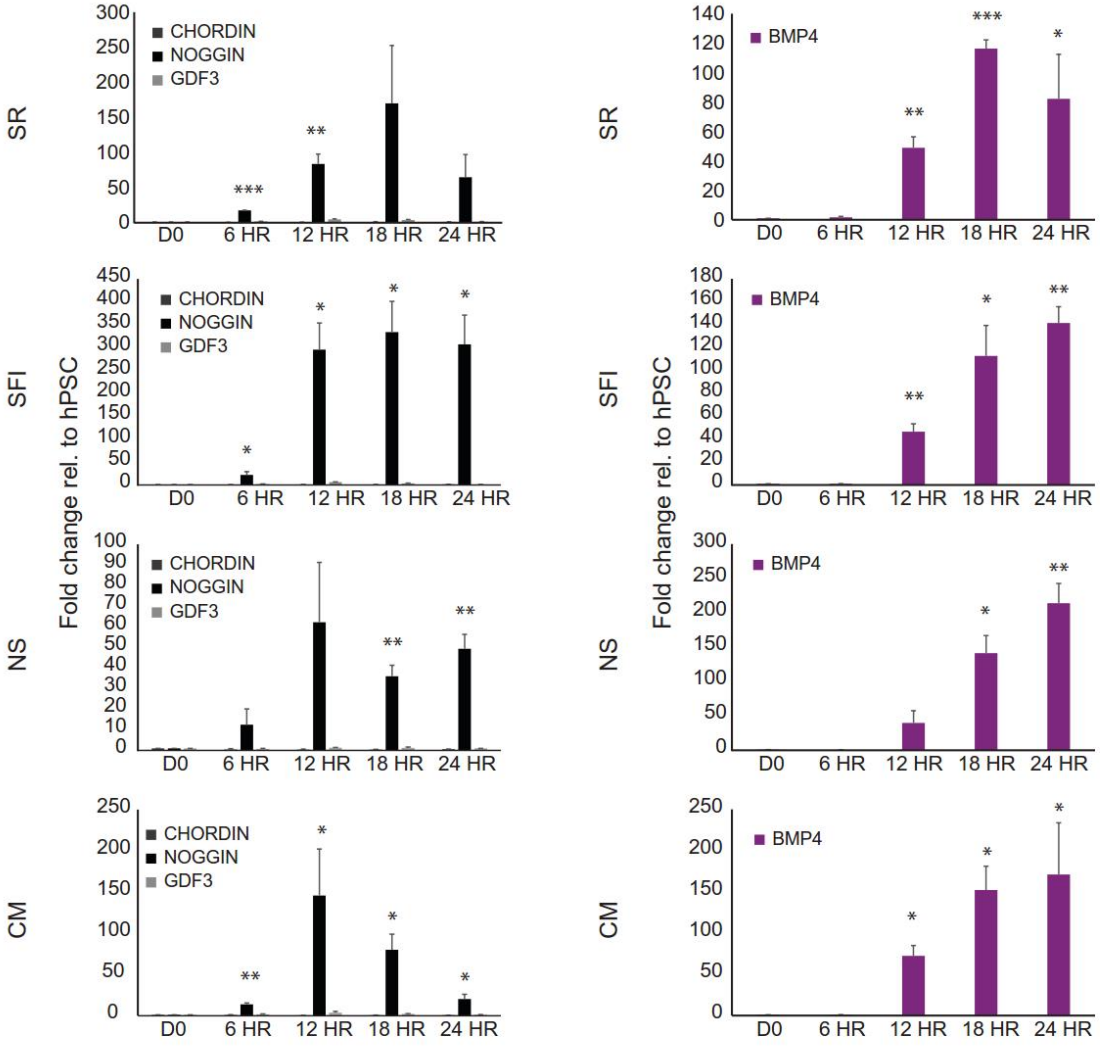


Figure S6: BMP4 induced upregulation of BMP4 and NOGGIN in tested medium conditions. Kinetic gene expression profiles for BMP4 and its cardinal inhibitors in response to BMP4 induced differentiation. Medium conditions tested include a Knockout serum – based medium (SR), a serum-free medium (SFI – for composition, please see Nazareth *et al*., Nature Methods 2013), Nutristem (NS), and Mouse Embryonic Fibroblast conditioned medium (CM). Data represented as mean and S.D of three biological replicates. * p<0.05, ** p<0.01, *** p<0.001.

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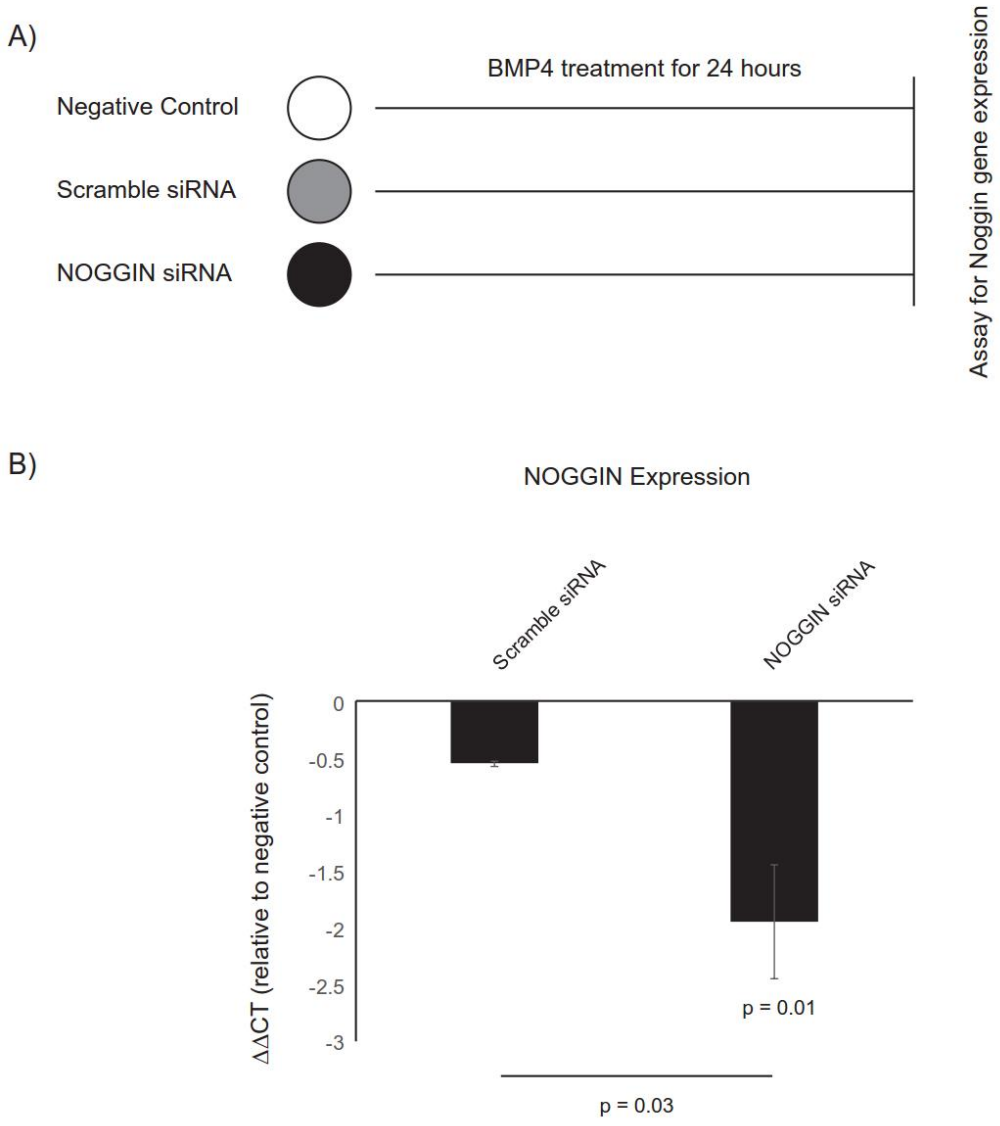


Figure S7: Controls for NOGGIN and Scramble siRNA. A) Overview of experimental setup. hPSC cultures were treated with Scramble, and NOGGIN siRNA with BMP4 for 24h. B) NOGGIN gene expression for the Scramble and NOGGIN siRNA relative to the negative control shown as ΔΔCT. Data represented as mean (s.d.) of three biological replicates. The p values were calculated using Student's t-test.

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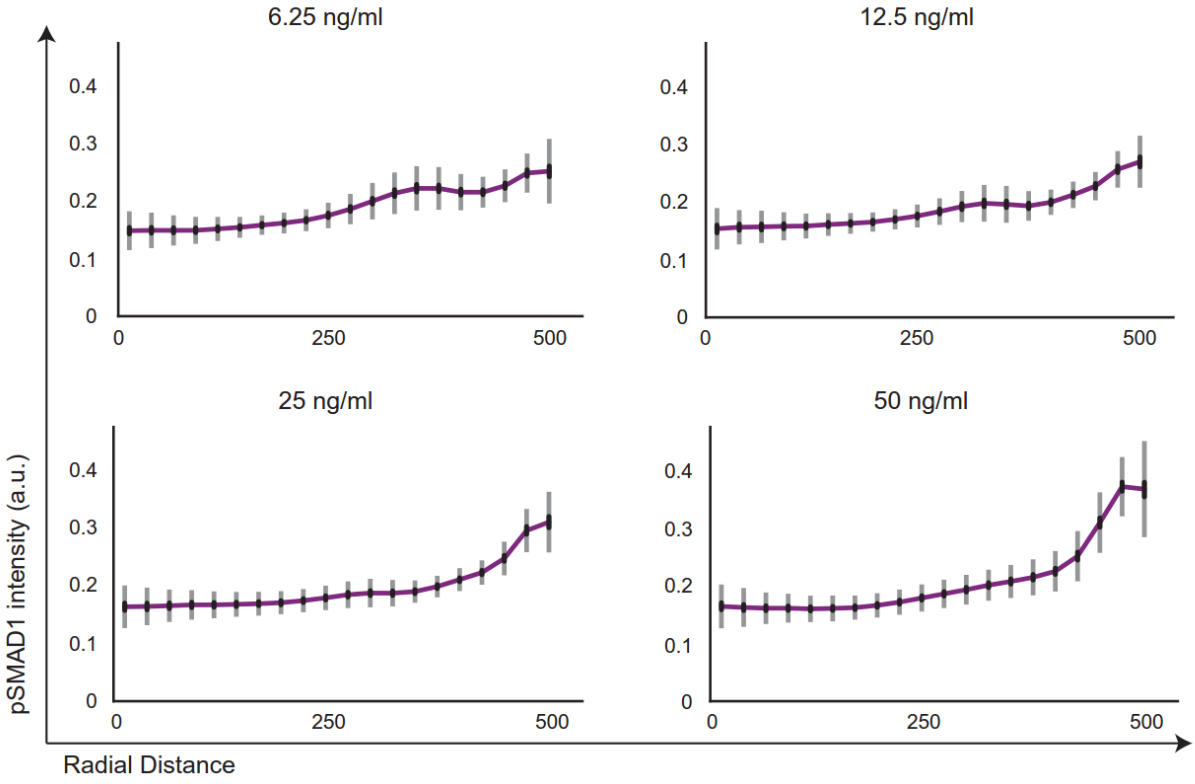


Figure S8: Quantified radial trends of pSMAD1 activity at 24 hours after induction with varying concentrations of BMP4. Radial trends of pSMAD1 activity were observed in varying BMPi concentrations (6.25 ng/ml, 12.5 ng/ml, 25 ng/ml, and 50ng/ml). Standard deviations shown in grey and 95% confidence intervals shown in black.

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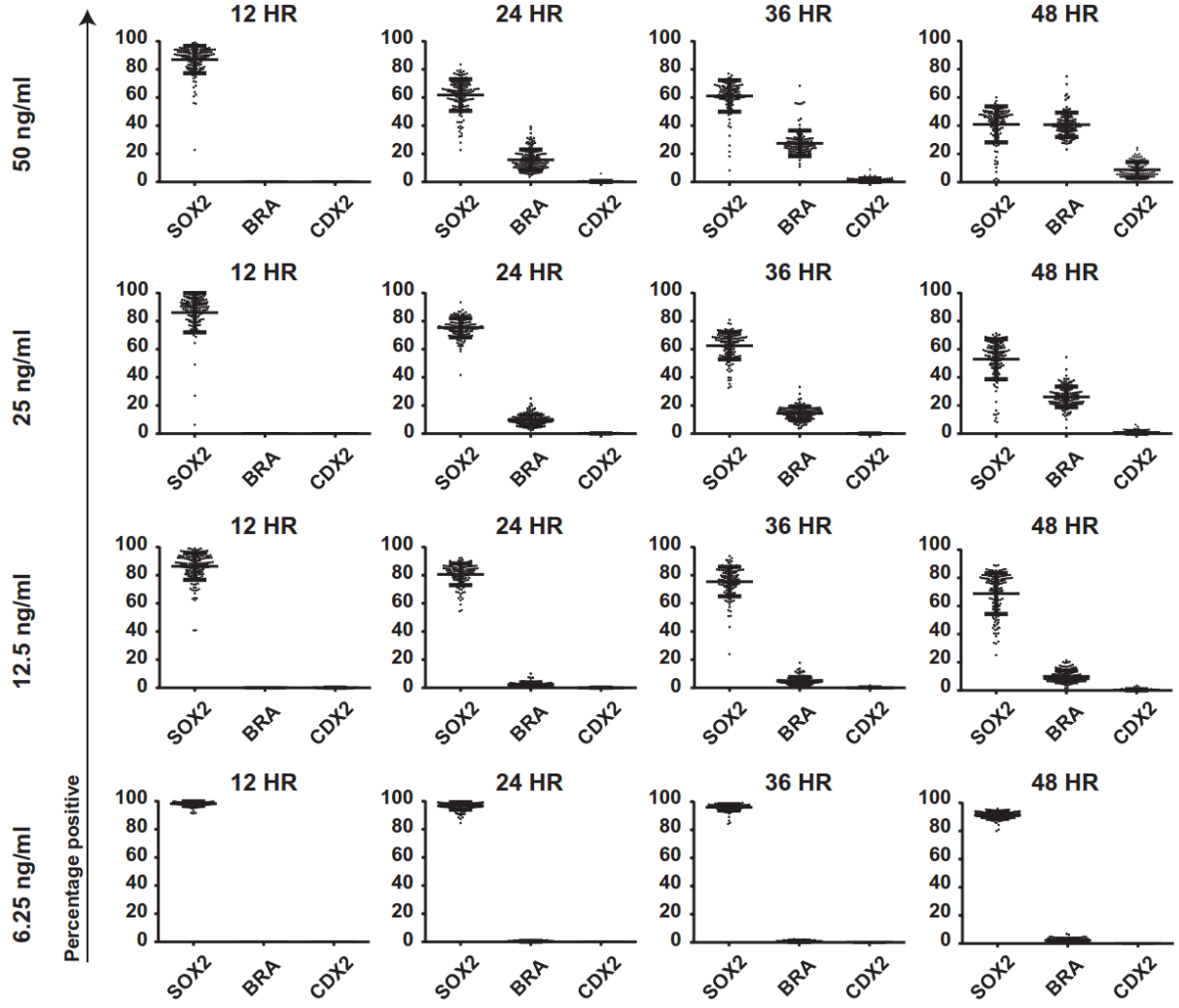


Figure S9: CDX2 and BRA expression in colonies arise as a function of BMP4 dose, and induction time. Percentage of cells expressing SOX2, BRA, and CDX2 in colonies induced to differentiate at varying concentrations of BMP4 (6.25 ng/ml, 12.5 ng/ml, 25 ng/ml, and 50 ng/ml) and induction times (12 hours, 24 hours, 36 hours, and 48 hours). Each condition had over 140 colonies. Data pooled from two experiments.

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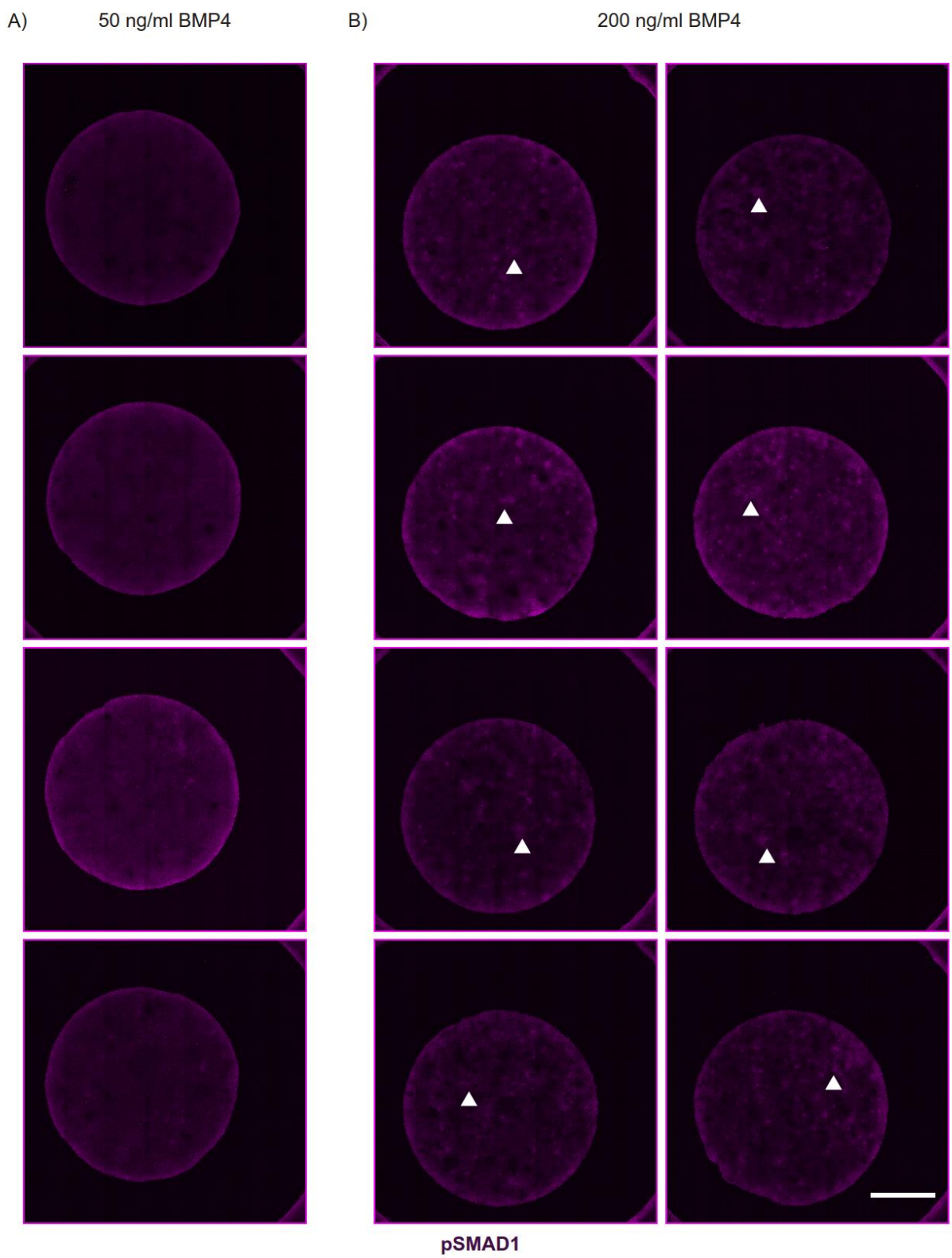


Figure S10: RD-like patterns noted in pSMAD1 activity in 3mm colonies when differentiated with high doses of BMP4. Representative immunofluorescent images of 3mm

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diameter colonies stained for pSMAD1 for a BMP4 dose of A) 50ng/ml, and B) 200ng/ml. White arrowheads denote representative areas of high pSMAD1 activity indicative of RD-like patterns. Scale bars represent 1mm.

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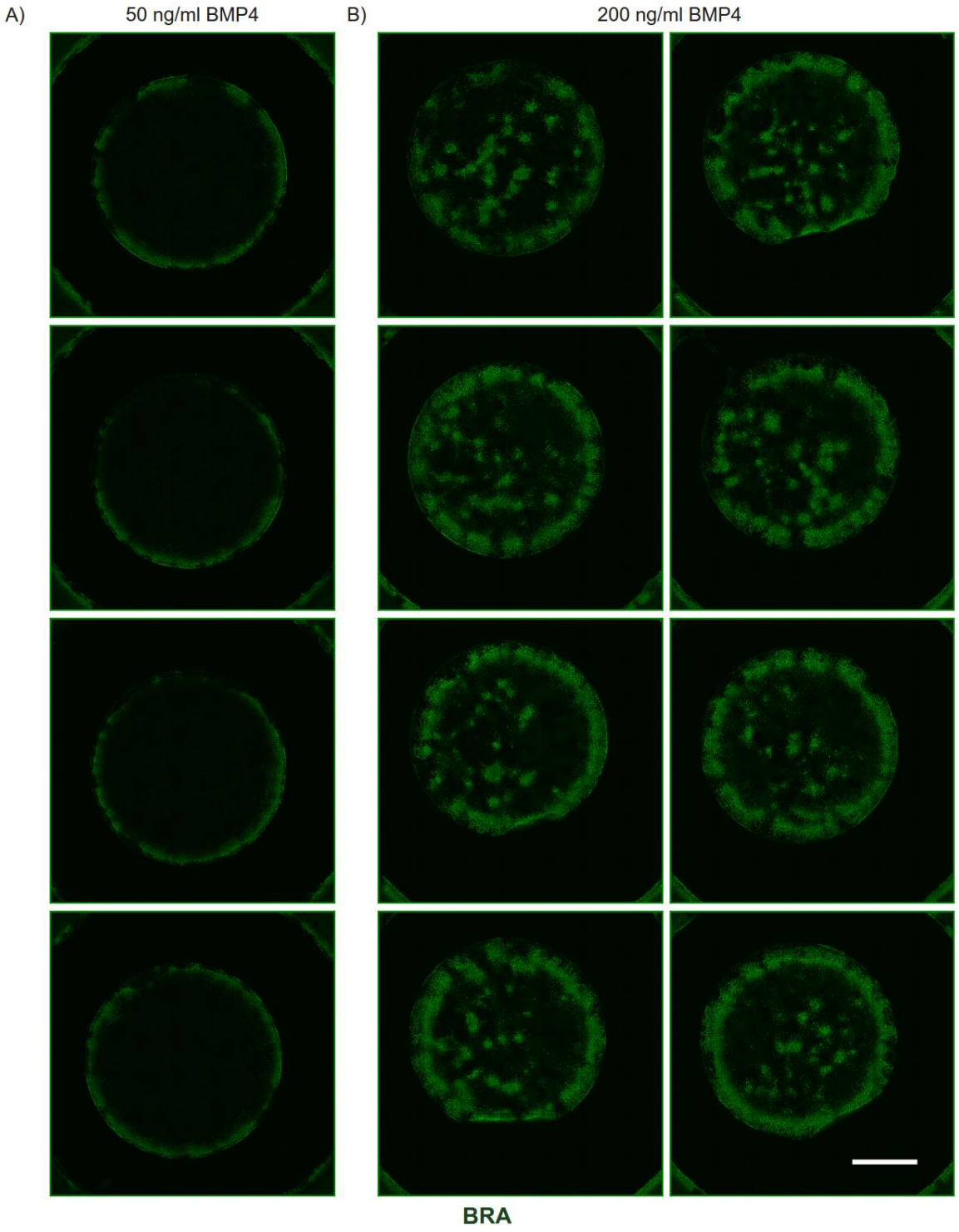


Figure S11: RD-like patterns noted in BRA fate acquisition in 3mm colonies differentiated in high doses of BMP4. Representative immunofluorescent images of 3mm diameter colonies stained for BRA for a BMP4 dose of A) 50ng/ml, and B) 200ng/ml. Scale bar represents 1mm.

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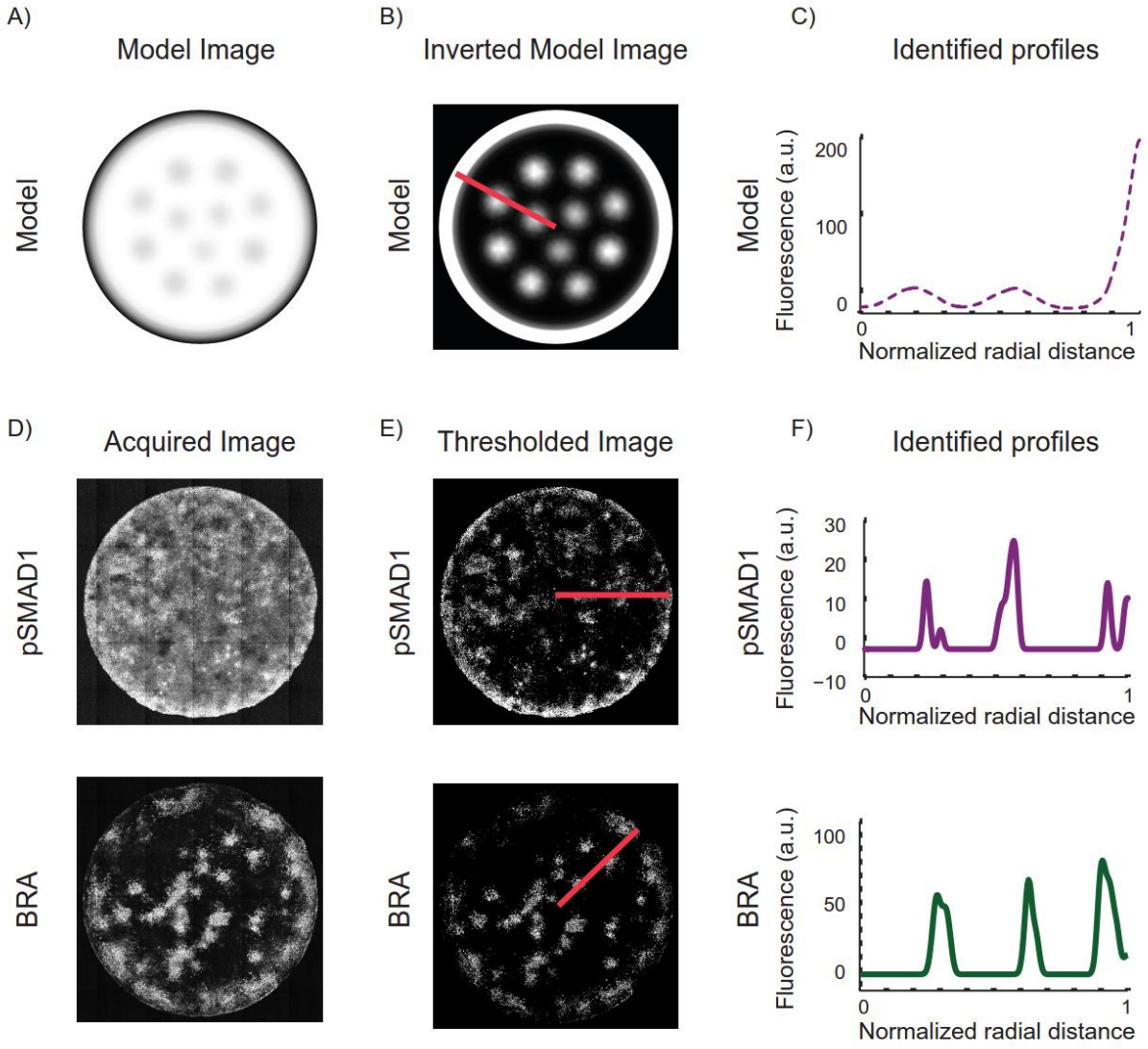
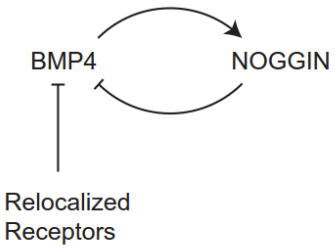


Figure S12: Analysis pipeline for extracting dominant periods of theoretically predicted distribution of free BMP4 ligands and expression patterns in experimental colonies. A) Model prediction, and the inverted image (B) shown in grey scale. Profiles extracted every 30 degrees. C) Profile observed along red line in B. Acquired (D), and thresholded (E) (to remove background noise) images for pSMAD1, and BRA. Profiles extracted every 30 degrees for multiple colonies (n = 28 for pSMAD1, and n=47 for BRA). F) Identified periodic profile along the red lines in (E) for pSMAD1 and BRA.

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A)



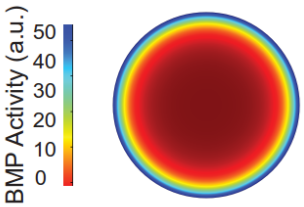
B)

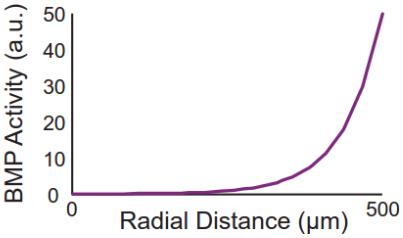
$$\frac{\partial bmp}{\partial t} = -k_1 nog - k_2 - k_3 bmp + D_{BMP}\nabla^2 bmp$$

$$\frac{\partial nog}{\partial t} = k_4 bmp - k_5 nog + D_{NOG}\nabla^2 nog$$

All coefficients (k1, k2, k3, k4, k5 DBMP, and DNOG) are positive.
k_1 – inhibition of BMP signaling mediated by NOGGIN.
k_2 – Inhibition of BMP signaling mediated by relocalized receptors.
k_3 – Degradation rate of BMP4.
k_4 – Production rate of NOGGIN mediated by BMP signaling.
k_5 – Degradation rate of NOGGIN.
D_{BMP}, and D_{NOG} – Diffusivities of BMP4 and NOGGIN.

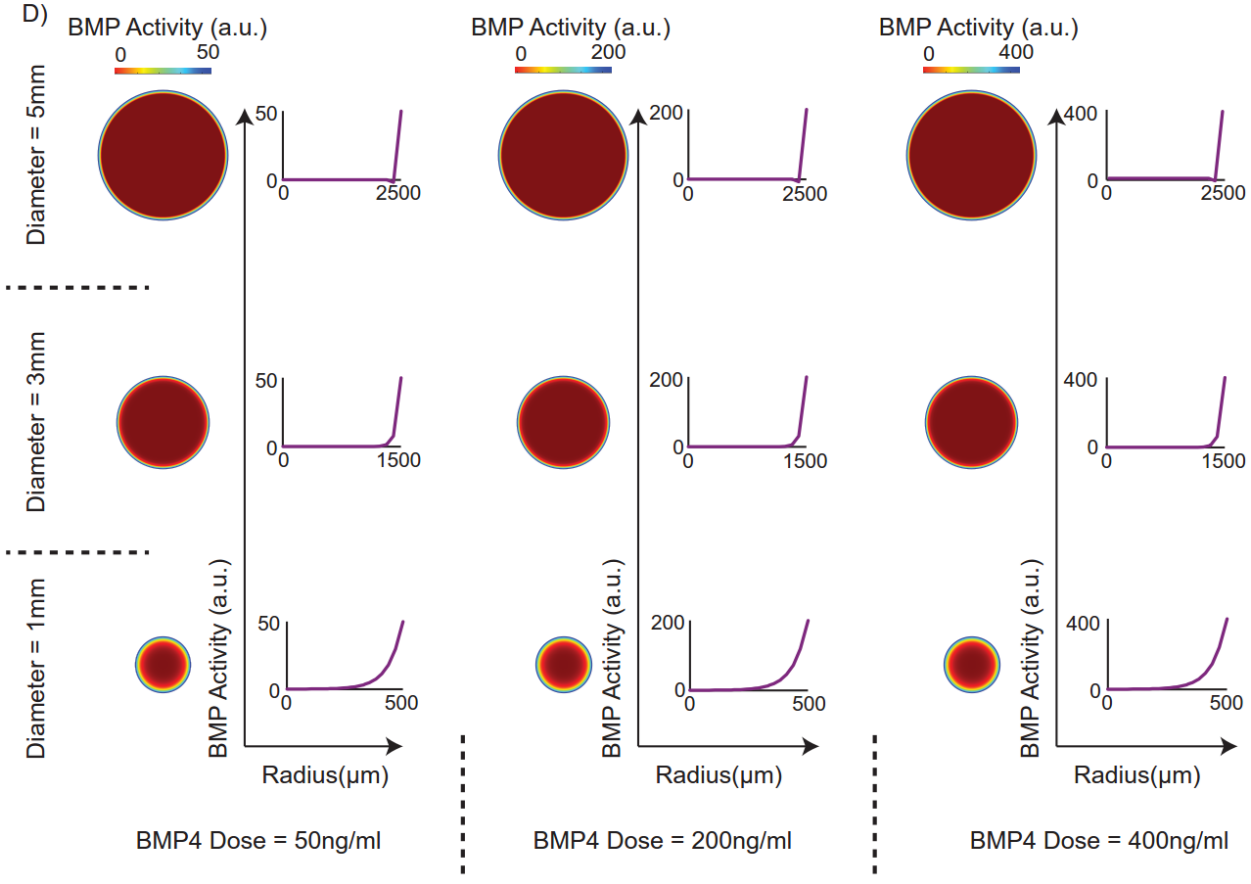
C)





k_1 = 0.01 [1/s]
k_2 = .003 [1/s]
k_3 = .003 [1/s]
k_4 = .008 [1/s]
k_5 = .009 [1/s]
D_{BMP} = 10μm²/s
D_{NOG} = 50μm²/s

D)



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Figure S13: Dual inhibition model does not give rise to repetitive RD-like free BMP4 distribution. A) Proposed dual-inhibition model of gradient formation in differentiating hPSC colonies by Etoc *et al*(Etoc et al. 2016). B) Simplified mathematical representation of a generic dual inhibition model. C) Gradient formation of free BMP4 ligands as predicted by the model. D) Varying doses and colony sizes demonstrates the inability of the dual-inhibition model to generate a periodic Turing-like response in the free BMP4 distribution.

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Supplement: Model Description

Summary of data in main text and motivation of the model

We observed that when geometrically-confined human pluripotent stem cell (hPSC) colonies were induced to differentiate in media supplemented with BMP4, pSMAD1 activity within the colonies self-organized into a radial gradient. We hypothesized that regulators of the BMP signaling pathway were controlling this observation and investigated the expression of key genes involved in the BMP signaling. We found that upon induction with BMP4, hPSCs upregulated the expression of both BMP4, and NOGGIN. Furthermore, siRNA mediated knockdown of NOGGIN interfered with the formation of the pSMAD1 gradient. These observations implicated a possible 'Activator/Inhibitor' model of reaction-diffusion (RD) mediated self-organization of signaling activity. Although BMP4 binds to its receptors with high affinity (K_d 10^{-9} – 10^{-10} M) (Zimmerman et al. 1996; Iwasaki et al. 1995), the affinity of NOGGIN and BMP4 has been calculated (Kd ~ 19 pM) (Zimmerman et al. 1996) and shown to be sufficiently high to prevent the BMP ligands' interaction with cognate receptors. Furthermore, there is evidence that NOGGIN is a diffusible molecule (Inomata et al. 2013) that is capable to inducing a long-range inhibition of BMP signaling. Therefore, they make a stereotypical activator/inhibitor pair.

To test the assertion that a BMP4-NOGGIN RD system was self-organizing the pSMAD1 gradient, we set out to develop an RD model specific to our tissue geometry, initial, and boundary conditions to query testable predictions of signaling gradient formation that may arise in our geometrically confined hPSC colonies. Notably, the purpose of this model was not to attain detailed information of the response of the signal transduction within the differentiating cells, or identify the gene regulatory network that would allow differential induction of fates in response to different signaling levels. Instead, we set out to identify the simplest possible RD model that could produce a BMP signaling gradient for a 1000µm diameter colony when differentiated with 50ng/ml of BMP4 in the induction media – reminiscent of the pSMAD1 signaling patterns observed in our experimental data under those specific conditions. We then validated this model based on the predictions to perturbations of two key experimental parameters (**Fig. 4** – main text); and then employed this model to made testable predictions of the differentiation behavior of the hPSC colonies beyond the conditions under which the model was build (**Fig. 6, 7** - main text). The RD model described below is a simplified, and an idealized model. Nevertheless, this model generates predictions of morphogen (BMP4) distribution in response to perturbations of experimental conditions which is constantly shown to be consistent with both pSMAD1 gradient formation and associated hPSC differentiation.

Two-component Reaction-Diffusion system

We set out to develop a mathematical model of the concentration profiles of BMP4 and NOGGIN molecules as a function of space and time in our micro-patterned colonies, based on a reaction-diffusion (RD) system (Turing 1952; Murray 2008; Gierer & Meinhardt 1972). The partial differentiation equation (PDE) set can be described as follows:

$$\frac{\partial bmp}{dt} = F(bmp, nog) - d_{BMP} bmp + D_{BMP} \nabla^2 bmp$$

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$$\frac{\partial nog}{dt} = G(bmp, nog) - d_{NOG} nog + D_{NOG} \nabla^2 nog \quad (1)$$

Here, *bmp*, and *nog* are functions of both space, and time that represent the local concentrations of BMP4 and NOGGIN molecules at a particular point in our micro-patterned colonies. *F (bmp, nog)*, and *G (bmp, nog)* represent the non-linear functions which describe the production rates of BMP4, and NOGGIN. The degradation rates of the molecules are given by d_{BMP}, and $d_{NOG;}$ and D_{BMP}, and D_{NOG} represent the diffusivities of the molecules.

We assumed that the production terms of BMP4, and NOGGIN can be approximated by linear functions (close to the steady state). The nature of the production terms is such that as the values of *F*(*bmp,nog*), and *G*(*bmp*,*nog)* increase, the system transitions away from steady state, preventing convergence of the solutions(Murray 2008; Kondo & Miura 2010). Attempts to restrict the values for the morphogen near steady state, to enable convergence, have either used non-linear functions that saturate at increasing values (e.g. the Hill function(Sick et al. 2006)) or have enforced a range in which the linear approximation of the reaction function is confined(Kondo & Miura 2010). Since we used linear production functions (2) in our model (1), we chose the latter strategy and restricted the reaction functions to a defined range(Kondo & Miura 2010). The production terms are represented by:

$$F(bmp, nog) = 0 \leq a_{BMP} bmp + b_{BMP} nog + c_{BMP} \leq 1$$
$$G(bmp, nog) = 0 \leq a_{NOG} bmp + b_{NOG} nog + c_{NOG} \leq 5 \quad (2)$$

Changing Variables:

To circumvent the issue of intractability of the number of BMP4 and NOGGIN molecules in the circular region of interest modelled by our PDE solutions, we chose to change the variables *bmp*, and *nog* into normalized, dimensionless variables which we represent as *bmp**, and *nog**.

Quantities, and assumptions of note – All experiments were performed in micro-patterned 96-well plates with a volume of a ***100µl*** *of induction media per well. The culture surface area of each well is* ***0.3165cm²****. The molecular weight of BMP4 is* ***34KDa****. We assumed that the two-dimensional colony 'surface' across which the BMP4 and NOGGIN distributions are predicted in the PDE set could be approximately represented by a* ***1µm*** *height from the colony surface. Therefore, the equivalent 'surface concentration' of* ***one colony of 1mm diameter*** *when* ***100ul*** *of induction media containing* ***1ng/ml*** *of BMP4 in SI units is* ***1.77x10^{10} molecules/m²****. We opted to change the variables (bmp, nog) by normalizing the entire PDE set by 1.77x10^{10} molecules/m^2.*

The linearized PDE set from (1), and (2) together are of the following form:

$$\frac{\partial bmp}{\partial t} = a_{BMP}bmp - b_{BMP}nog + c_{BMP} - d_{BMP}bmp + D_{BMP}\nabla^2 bmp$$
$$\frac{\partial nog}{\partial t} = a_{NOG}bmp - b_{NOG}nog + c_{NOG} - d_{NOG}nog + D_{NOG}\nabla^2 nog \quad (3)$$

The SI units for the parameters, and variables in (3) are shown below:

$bmp\ [\frac{molecules}{m^2}]$	$nog\ [\frac{molecules}{m^2}]$
$a_{BMP}\ [\frac{1}{s}]$	$a_{NOG}\ [\frac{1}{s}]$
$b_{BMP}\ [\frac{1}{s}]$	$b_{NOG}\ [\frac{1}{s}]$
$c_{BMP}\ [\frac{molecules}{m^2 \times s}]$	$c_{NOG}\ [\frac{molecules}{m^2 \times s}]$
$d_{BMP}\ [\frac{1}{s}]$	$d_{NOG}\ [\frac{1}{s}]$
$D_{BMP}\ [\frac{m^2}{s}]$	$D_{NOG}\ [\frac{m^2}{s}]$

Dividing throughout by 1.77x10^{10} ***molecules/m²***, we changed the variables *bmp, nog,* c_{BMP}, and c_{NOG} as follows:

$bmp\ [\frac{molecules}{m^2}]$ = *bmp** [ ] x1.77x10^{10} $[\frac{molecules}{m^2}]$	$nog\ [\frac{molecules}{m^2}]$ = *nog** [ ] x1.77x10^{10} $[\frac{molecules}{m^2}]$
$c_{BMP}[\frac{molecules}{m^2 Xs}] = c^*_{BMP}\ [\frac{1}{s}]$ x1.77x10^{10} $[\frac{molecules}{m^2}]$	$c_{NOG}[\frac{molecules}{m^2 Xs}] = c^*_{NOG}\ [\frac{1}{s}]$ x1.77x10^{10} $[\frac{molecules}{m^2}]$

Taken together, the updated set of partial differential equations with the changed variables (*bmp**, *nog*,* c^*_{BMP} *and* c^*_{NOG}) is given below:

$$\frac{\partial bmp^*}{\partial t} = a_{BMP}bmp^* + b_{BMP}nog^* + c^*_{BMP} - d_{BMP}bmp^* + D_{BMP}\nabla^2 bmp^*$$
$$\frac{\partial nog^*}{\partial t} = a_{NOG}bmp^* + b_{NOG}nog^* + c^*_{NOG} - d_{NOG}nog^* + D_{BMP}\nabla^2 nog^* \quad (4)$$

Dose dependence in BMP4 and NOGGIN production

We observed a BMPi dose dependent production of both BMP4 and NOGGIN (**Fig S5**). To incorporate this response into our model we chose the following expressions for a_{BMP}, and a_{NOG}.

$$a_{BMP} = \alpha(1 + BMPi * \gamma_{BMP})$$
$$a_{NOG} = \alpha(1 + BMPi * \gamma_{NOG}) \quad (6)$$

Initial conditions of BMP4 and NOGGIN distributions in micro-patterned colonies:

BMP4

BMP4 is added in the differentiation medium, and presented to the colonies at a uniform dose. Therefore, we considered the initial concentration of BMP4 to be a constant value given by *BMPi*.

NOGGIN

The initial conditions for NOGGIN are more nuanced. Although NOGGIN is produced in response to BMP signaling in the cardinal 'Activator-Inhibitor' paradigm, BMP4 ***inhibition*** in the initial condition (at time t=0), can be achieved by a variety of different molecules (e.g. FOLLISTATIN (FST), CHORDIN, GDF3, and CERBERUS-Like (CERL) among others) in addition to NOGGIN(Wu & Hill 2009). Since we observed elevated basal expression of BMP signaling inhibitors like FST, GDF3, and CERL relative to NOGGIN in hPSCs during basal culture conditions (**Fig. S4B-G**), we opted to consider the spatial profile of a 'generic BMP inhibitor' as the initial condition for the RD paradigm.

To identify the specific spatial profile of a generic BMP inhibitor, we developed a simplified model of a passive diffusion-driven profile that would arise in a circular hPSC colony where each cell is a source of the secreted molecule. Over the course of the formation of a confluent hPSC colony, we assumed that the expression profile of a 'generic BMP inhibitor' would reach a steady state. To approximate this steady state spatial profile, we considered each cell (a point source of the inhibitor), evenly distributed within the colony (**Fig. S14A**), and assumed an infinite sink at a large distance from the colony (**Fig. S14B**). Simulation of a steady state response revealed a spatial profile that could broadly be approximated as an elliptical paraboloid (**Fig. S14C-D**). Accordingly, we considered the initial condition of NOGGIN, which at t=0 can be replaced by the effective contribution of all BMP inhibitors being expressed by the hPSCs in the micro-patterned colony, to be an elliptical paraboloid function (**Fig. S14E**). Notably the peak concentration chosen for the NOGGIN initial condition is arbitrary – and the patterning of free BMP4 distribution was found to be robust to the choice of the peak value.

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Boundary conditions for the BMP4-NOGGIN reaction-diffusion system in micro-patterned colonies

We assumed that the cells at the radial edge of the micro-patterned colonies were always subjected to the same concentration of BMP4 that is in the bulk medium, which parallels the 'edge-sensing' model that has been proposed by previous studies(Etoc et al. 2016). For the case of NOGGIN, we assumed a no flux boundary condition.

Final Reaction-Diffusion PDE

The final two-component PDE can be written as shown below:

$$\frac{\partial bmp^*}{\partial t} = a_{BMP} bmp^* + b_{BMP} nog^* + c^*_{BMP} - d_{BMP} bmp^* + D_{BMP} \nabla^2 bmp^*$$

$$\frac{\partial nog^*}{\partial t} = a_{NOG} bmp^* + b_{NOG} nog^* + c^*_{NOG} - d_{NOG} nog^* + D_{BMP} \nabla^2 nog^*$$

Initial Conditions

bmp(t=0) = BMPi*

$$nog^*(t=0) = \{0 \leq -\left(\frac{x}{R}\right)^2 - \left(\frac{y}{R}\right)^2 + 1\}$$

Boundary Conditions

bmp(boundary) = BMPi*

$$\frac{d(nog * (boundary))}{dt} = 0$$

In the initial condition for Noggin, *R* represents the colony radius.

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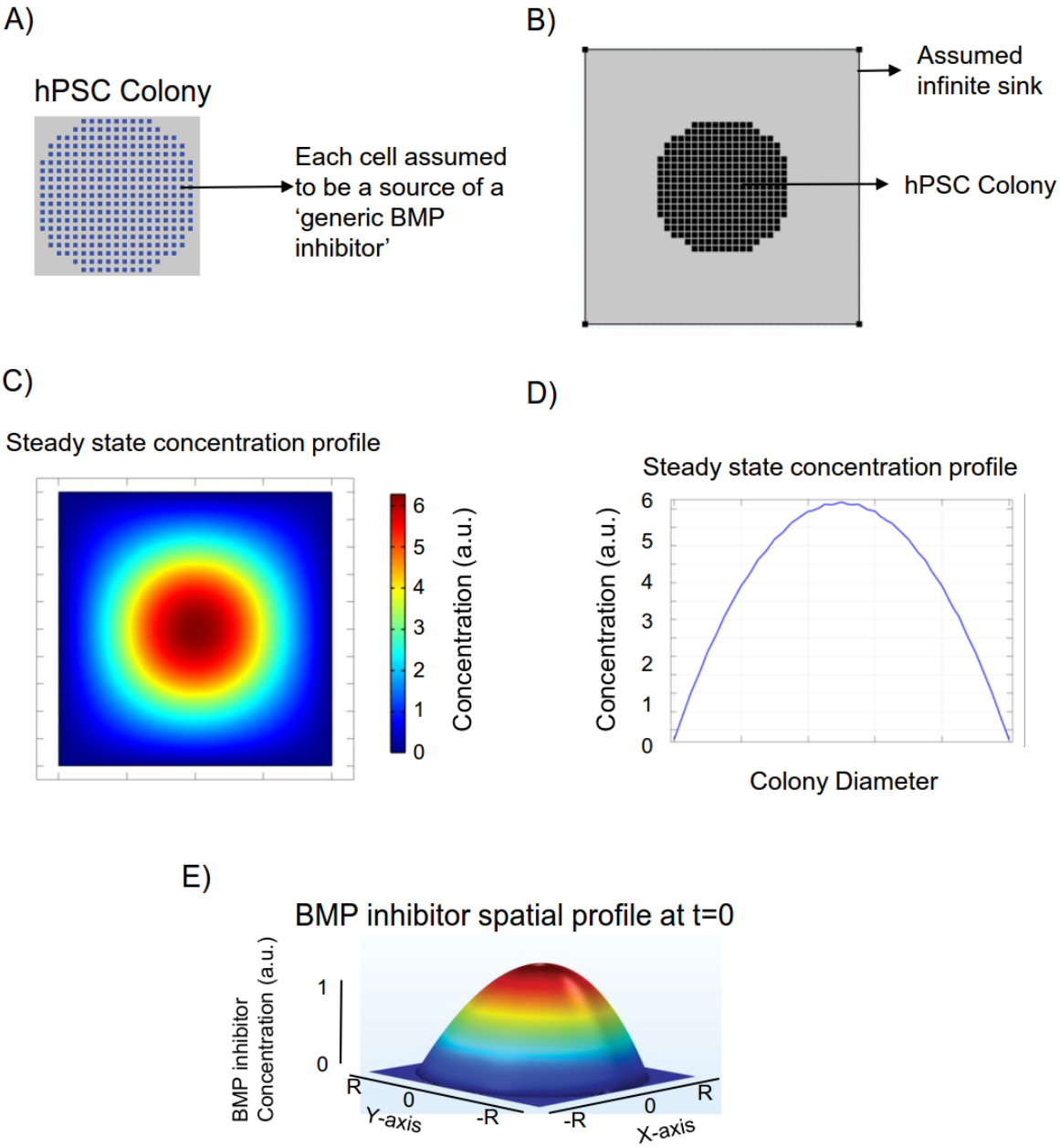


Figure S14: Initial condition for Noggin. A) A circular colony of cells (blue dots) modeled as a collection of sources of secreted molecules. B) Noggin is assumed to have an infinite sink at a certain distance from the colony periphery. The steady state diffusion profile of the secreted molecule is shown in C), and the expression profile within the colony along the diameter shown as a line plot D). E) The assumed initial condition of Noggin at the start of induction. 'R' represents the colony radius.

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Parameter choices and parameter sensitivity

Importantly, the coefficients ka_{BMP}, b_{BMP}, c^*_{BMP}, ka_{NOG}, b_{NOG}, and c^*_{NOG} in the production terms for BMP4, and NOGGIN do not correspond to experimentally determined parameter values. The values for these parameters were chosen as per Kondo et al(Kondo & Miura 2010) – we first chose a parameter set that resulted in oscillating values of BMP4, and NOGGIN, and then chose the values of the diffusion coefficients such that $D_{NOG} > D_{BMP4}$ (**Table S3**).

The diffusivity values for both NOGGIN, and BMP4 are in realistic ranges (Raspopovic et al. 2014; Sick et al. 2006; Inomata et al. 2013). For instance, Inomata et al. calculated the diffusivity of NOGGIN in Xenopus embryos to be 37±6.6 $\mu m^2/s$, which is close to the predicted diffusivity of Noggin in our system. However, the exact values of the diffusivities in our system have not been measured.

The value for b_{NOG} was chosen to be zero since NOGGIN does not have any receptors and is therefore, unable to repress its own expression.

Table S3: Model parameters

Parameter	Value	Adapted from
α	0.005 [1/s]	Kondo et al.
b_{BMP}	0.01[1/s]	Kondo et al.
c^*_{BMP}	0.003 [1/s]	Kondo et al.
d_{BMP}	0.003 [1/s]	Kondo et al.
D_{BMP}	11 [µm2/s]	-
γ_{NOG}	0.0025	Kondo et al.
b_{NOG}	0	-
c^*_{NOG}	-0.015 [1/s]	Kondo et al.
d_{NOG}	0.009 [1s]	Kondo et al.
D_{NOG}	55 [µm2/s]	-

In **Fig. S15**, we show the response of the predicted spatial profile of free BMP4 molecules within the geometrically-confined hPSC colony to varying the above parameters to provide a sense of the sensitivity of the model output to the model parameters. The sensitivity of the model to perturbing the mesh definition is shown in **Fig. S16**.

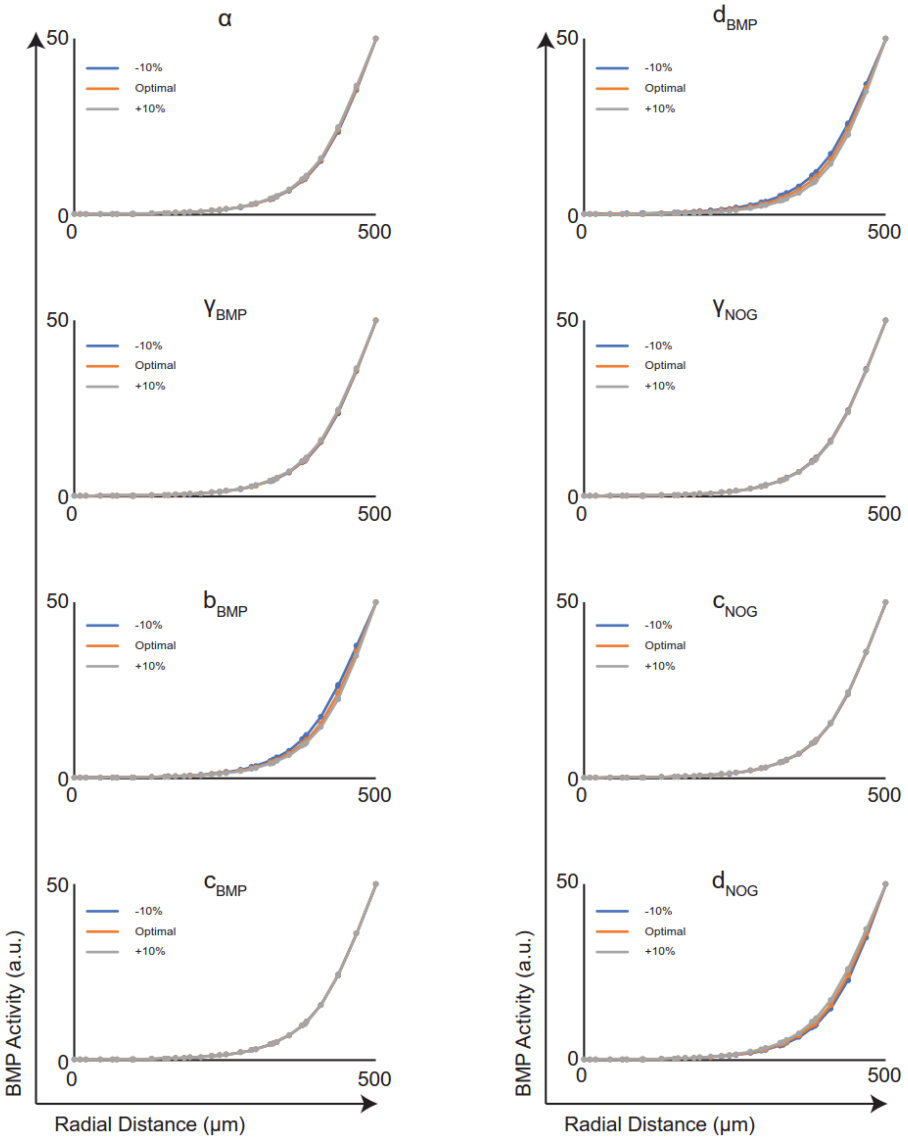


Figure S15: *Response of predicted gradient formation to perturbation of model parameters.*

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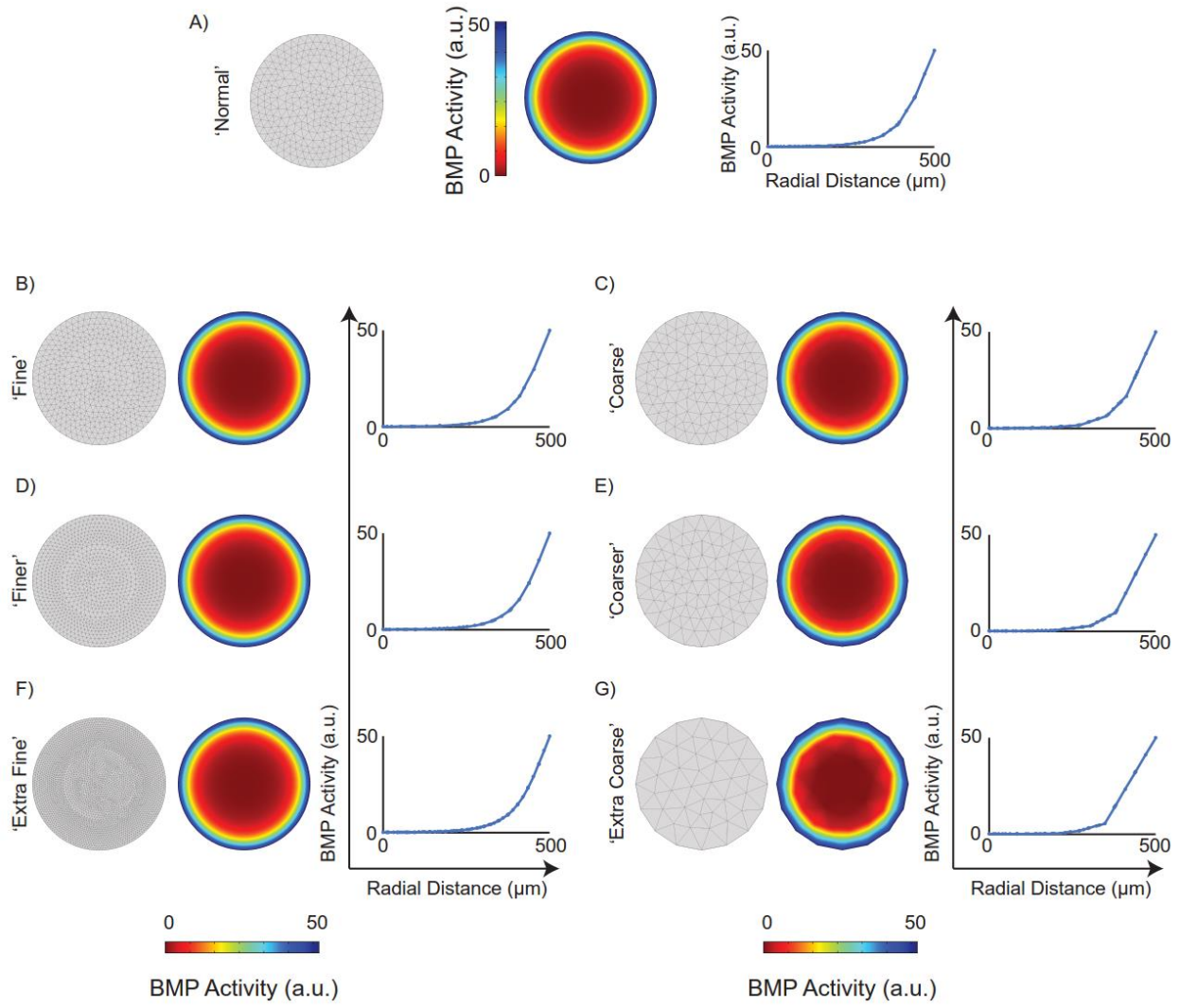


Figure S16: *Gradient formation predicted with pre-defined mesh sizes in Comsol A) 'Normal', B) 'Fine', C) 'Coarse', D) 'Finer', E) 'Coarser', F) 'Extra Fine', and G) 'Extra Coarse'.*

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